

Predicting sleep apnea from polysomnography

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github.com/dertmatt24/projectappliedstatistics2026

1. Problem Framing

Polysomnography

n. Multi-parameter type of sleep study and a diagnostic tool in sleep medicine.

Sleep Apnea

n. Sleep-related breathing disorder in which repetitive pauses in breathing during sleep result in poor ventilation and sleep disruption.

Can we predict sleep apnea attacks from polysomnography with EEG alongside purely physiological per-patient features?

→ Should we approach this as a *pure classification* problem or as a *regression* problem?

Regression Vs Classification

Option 1: Direct Classification

→ Train the model to predict the severity class directly.

Pros:

- Simple, standard pipeline
- Loss function is aligned with the clinical outcome

Cons:

- Throws away information: a prediction of 4.9 and 5.1 are treated as maximally wrong
- Class imbalance is likely and needs to be handled explicitly

Option 2: Regression + Post-hoc Binning

→ Train the model to predict AHI then apply thresholds.

Pros:

- Preserves the natural shape of the target variable from the dataset
- Errors near boundaries are penalized proportionally, not catastrophically

Cons:

- The model is not directly optimized for classification accuracy

2. Dataset Presentation & Study

<https://www.kaggle.com/datasets/yfrite/polysom>

Features & Structure

Biometric Data:

- User ID: A unique id for 40 patients
- Night ID: A unique id for each of the 2 nights
- Age: [years]
- Sex: [F: female, M: male]
- Height: [Centimeters]
- Weight: [Kilograms]
- Pulse: [Beats per minute]
- BPsyst/ BPdia: systolic blood pressure/ diastolic blood pressure of the patient
- ODI: Oxygen Desaturation index of the patient for the night

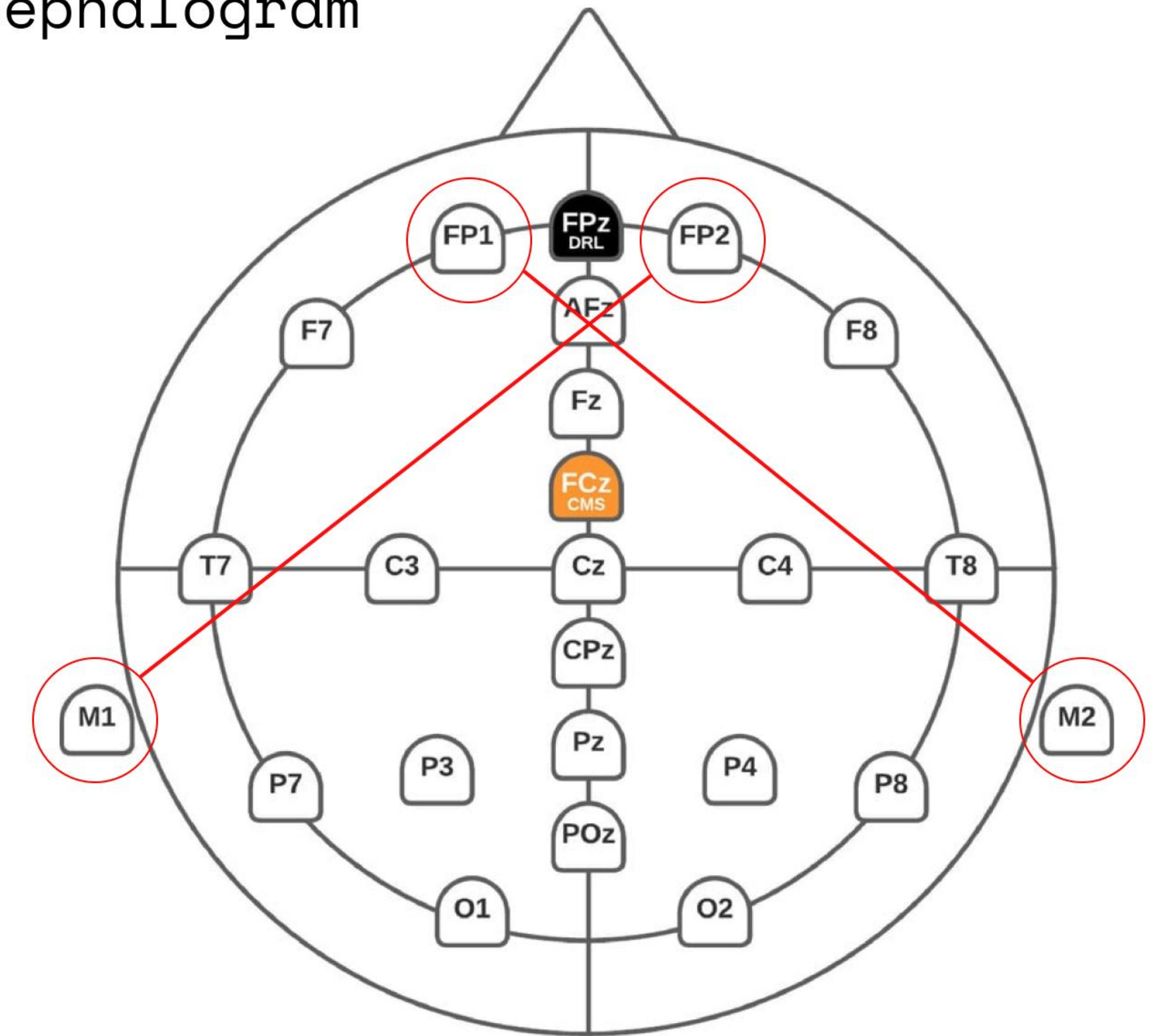
Diagnostic Data:

- NAp: number of cases apnoe
- NHyp: number of cases hyperapnoe
- AI: Apnoe Index = $\text{NAp} / \text{TimeOfRecordInHours}$
- HI: Hyperapnoe Index = $\text{NHyp} / \text{TimeOfRecordInHours}$
- AHI: Apnoe-Hyperapnoe Index = $(\text{NAp} + \text{NHyp}) / \text{TimeOfRecordInHours}$

Electroencephalogram

Channels:

- Fp1-M2
- C3-M2
- O1-M2
- Fp2-M1
- C4-M1
- O2-M1



Issue & Caveats

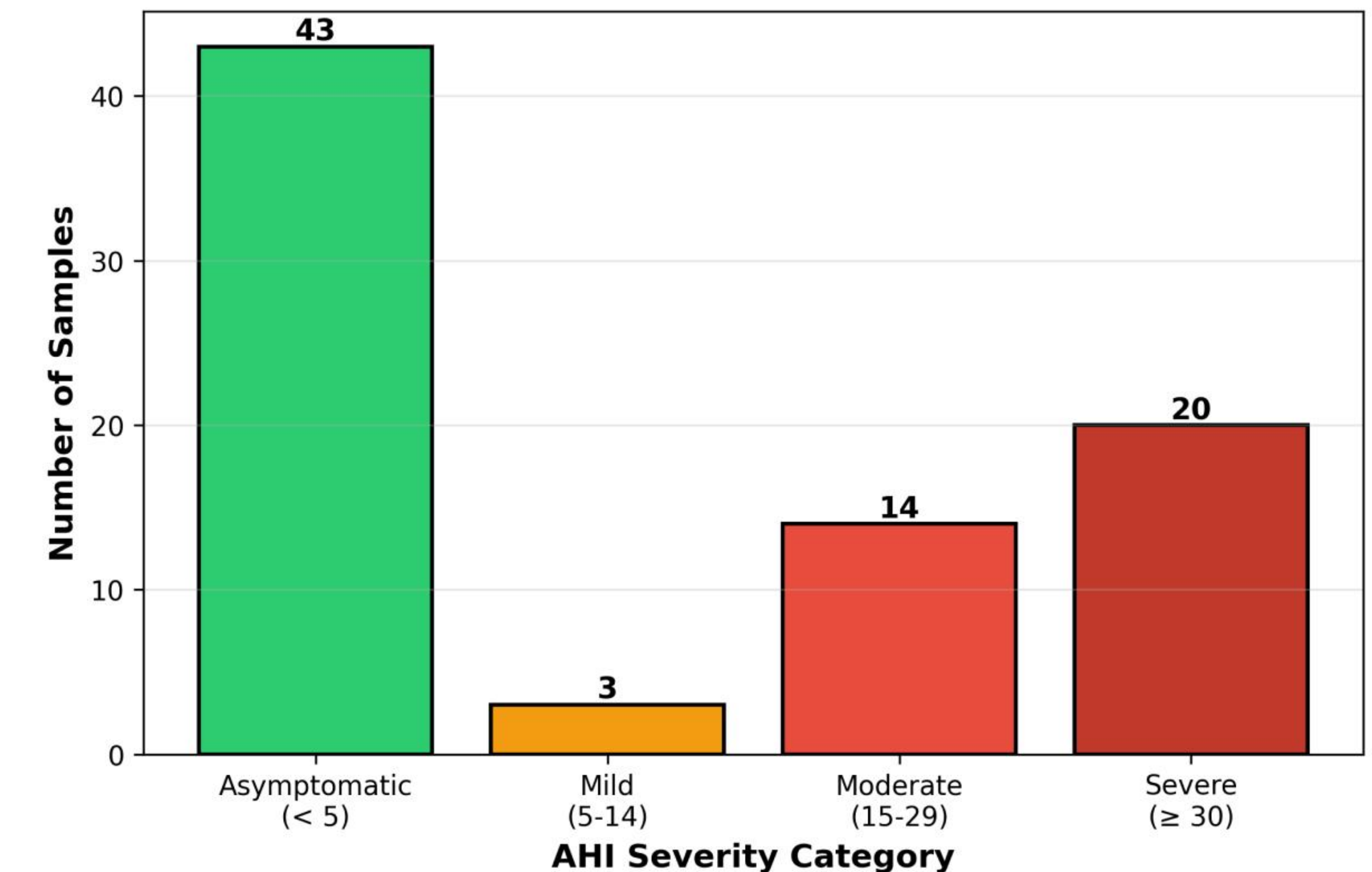
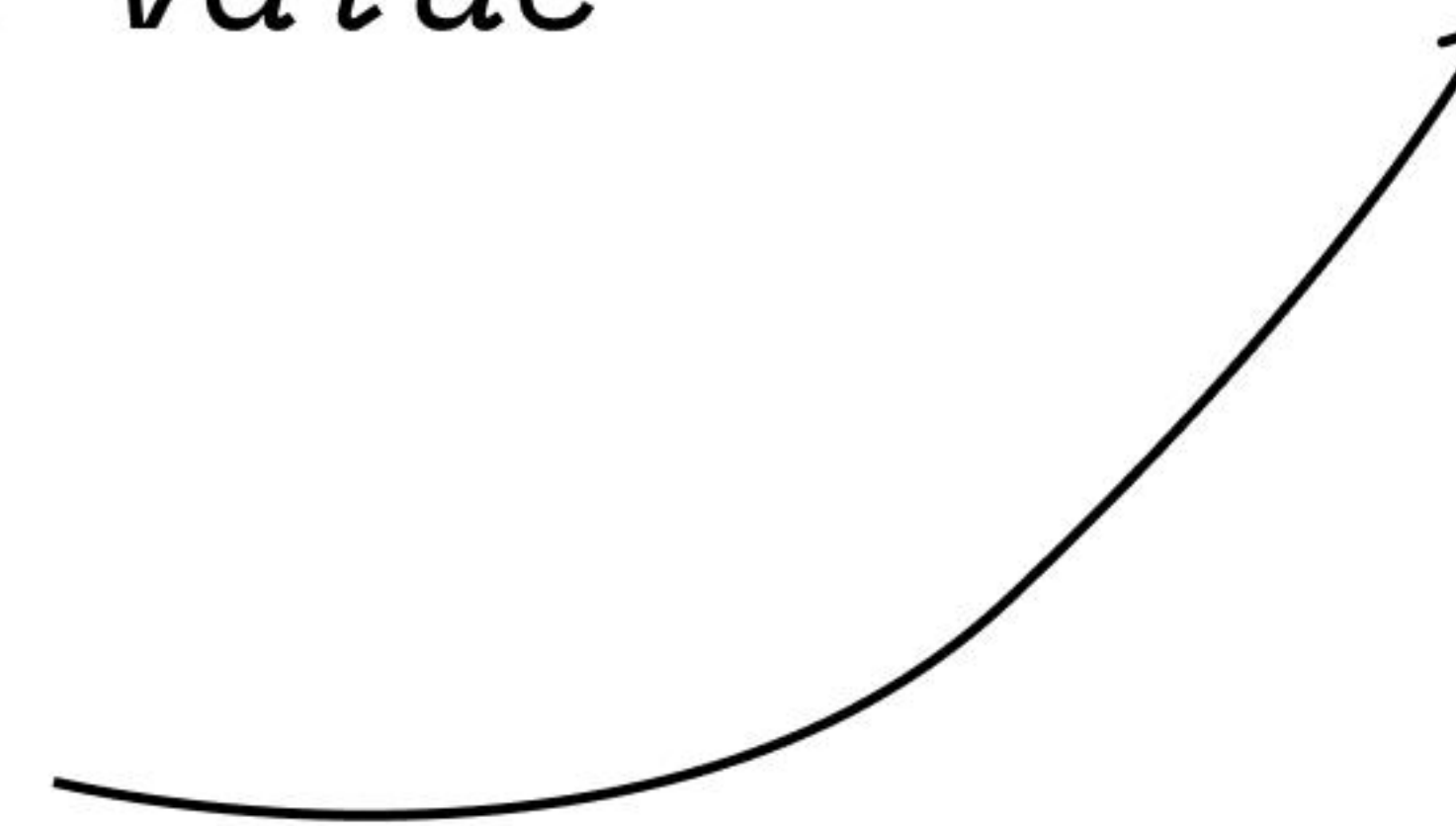
n = 80 (2 deleted, heavy artifact channels)

→ DATA SCARSITY: Labeled datapoints are hard to obtain, manual labeling and patient observation are required

- **Healthy patients have exactly 0.0 AHI**

→ More on this later, imbalance towards a value

- **Class imbalance towards the two extremes (no symptoms, heavy symptoms)**



- **Average recording length: 8hours**

→ Usual literature windows: 60 seconds!

*8hours*200hz = c.a. 6.000.000 samples per label, extremely high dimensional landscape*

- **We need to find a way to exploit the multimodal datapoints: biometric data & EEG recordings together**

Also, multiple recordings per-patient! (more on this later)

Class Imbalance & Partitioning

Problem: Data Leakage via Patient Overlap

→ The model learns patient-specific quirks, not generalizable patterns, overestimating the model performance.

→ Solution: patient-level splits, all the datapoints for a patient are either training or test.

Problem: Class imbalance leads to the majority class baseline trap

→ Predicting the major class leads an expected 55% accuracy

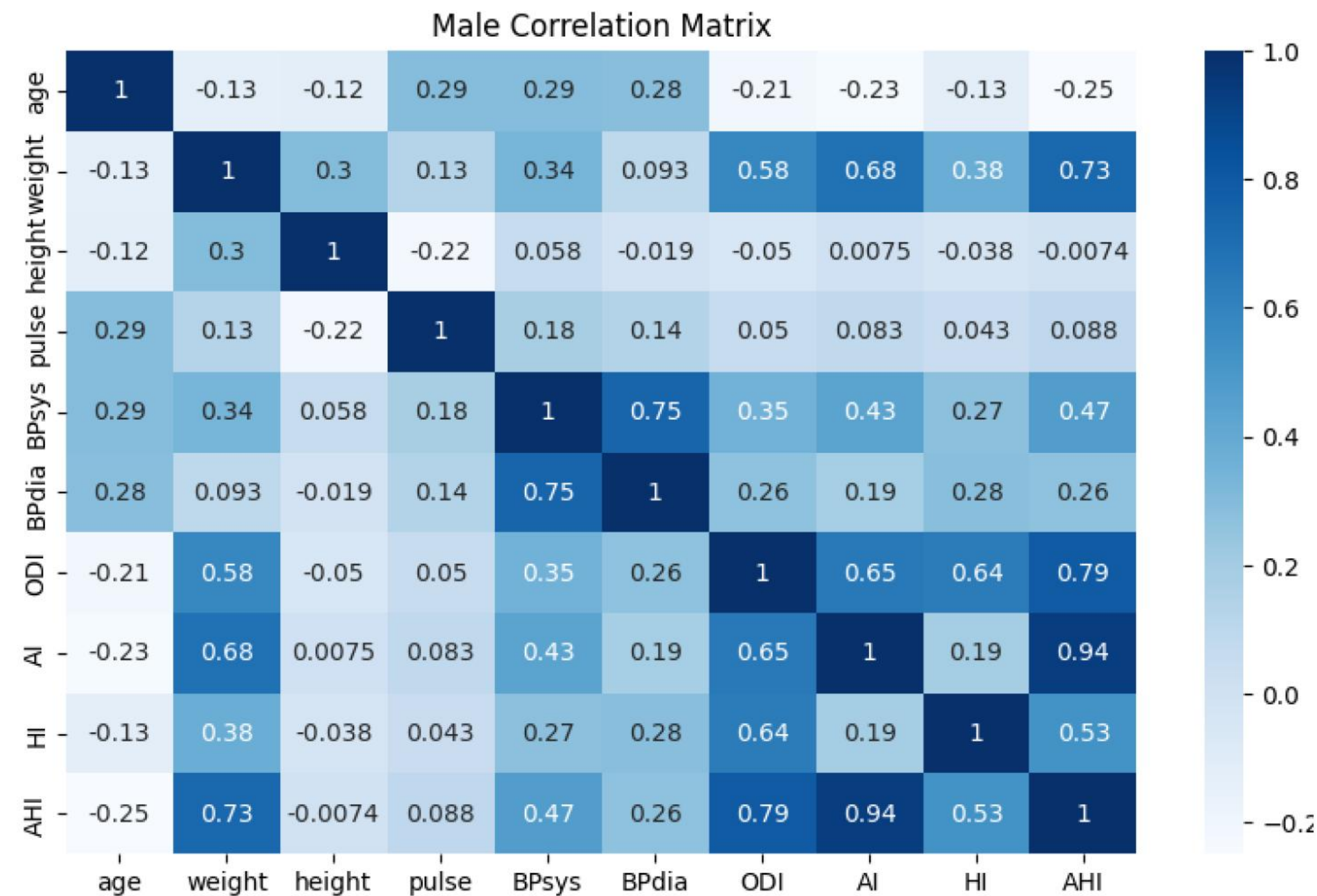
→ Naive sampling will ignore less probable classes.

→ Solution: Sample with prior class probabilities (stratification)

Can we solve alleviate class imbalance problem?

- ★ • Drop altogether the “Mild” class
- Tweak the thresholds a bit (not physiologically meaningful)

Multicollinearity

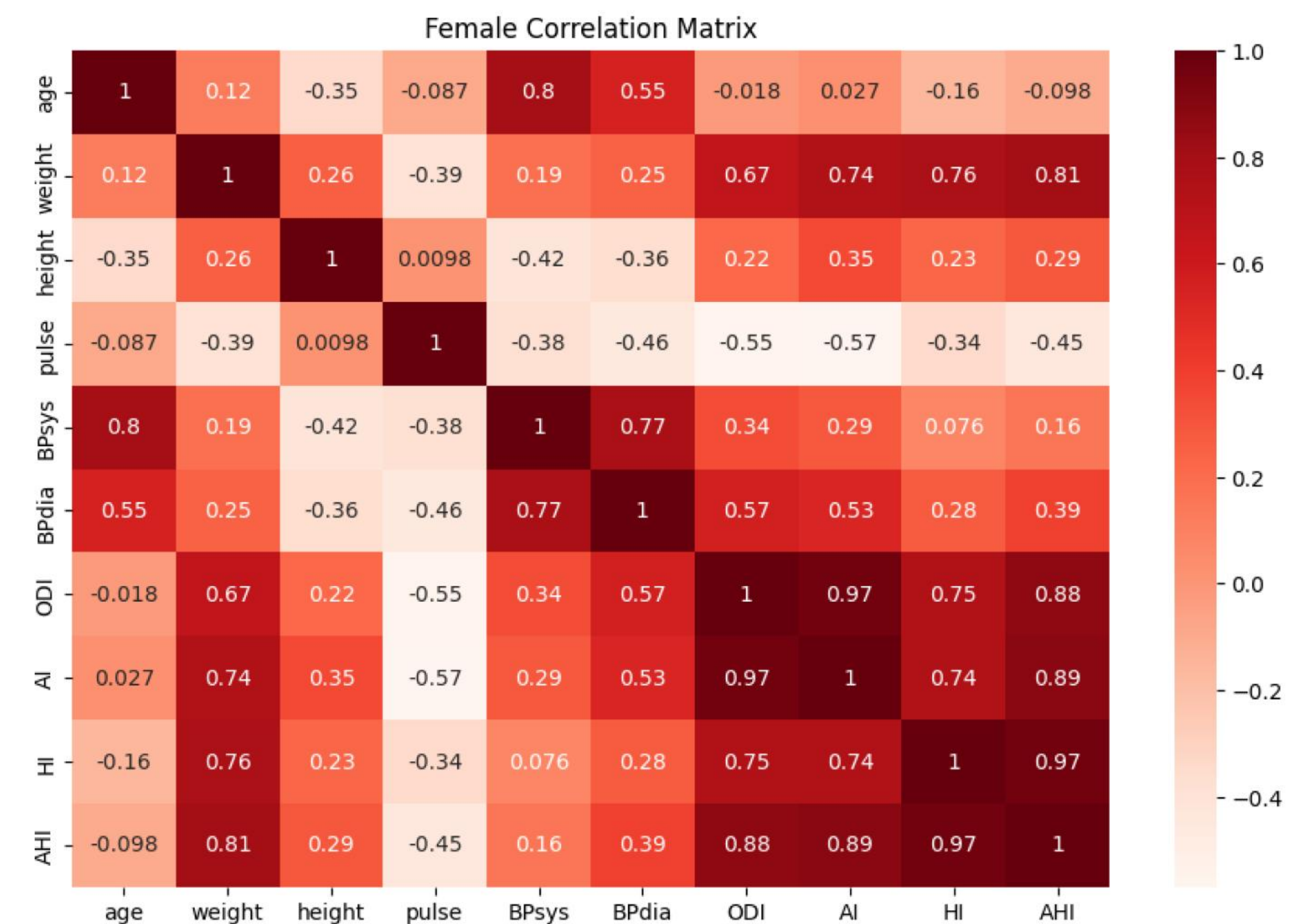


Male:

- “weight”: “ODI” and “BPsys”
- “age”: “BP” and “pulse”
- “height”: “weight”
- “BPsys”: “BPdia”
- “ODI”: “BPsys”

AHI Predictors:

- “ODI”, “weight”, “BPsys”



Female:

- “age”: “BPsys” and “BPdia”
- “weight”: “ODI”
- “BPsys”: “BPdia”
- “ODI”: “BPsys” and “BPdia”

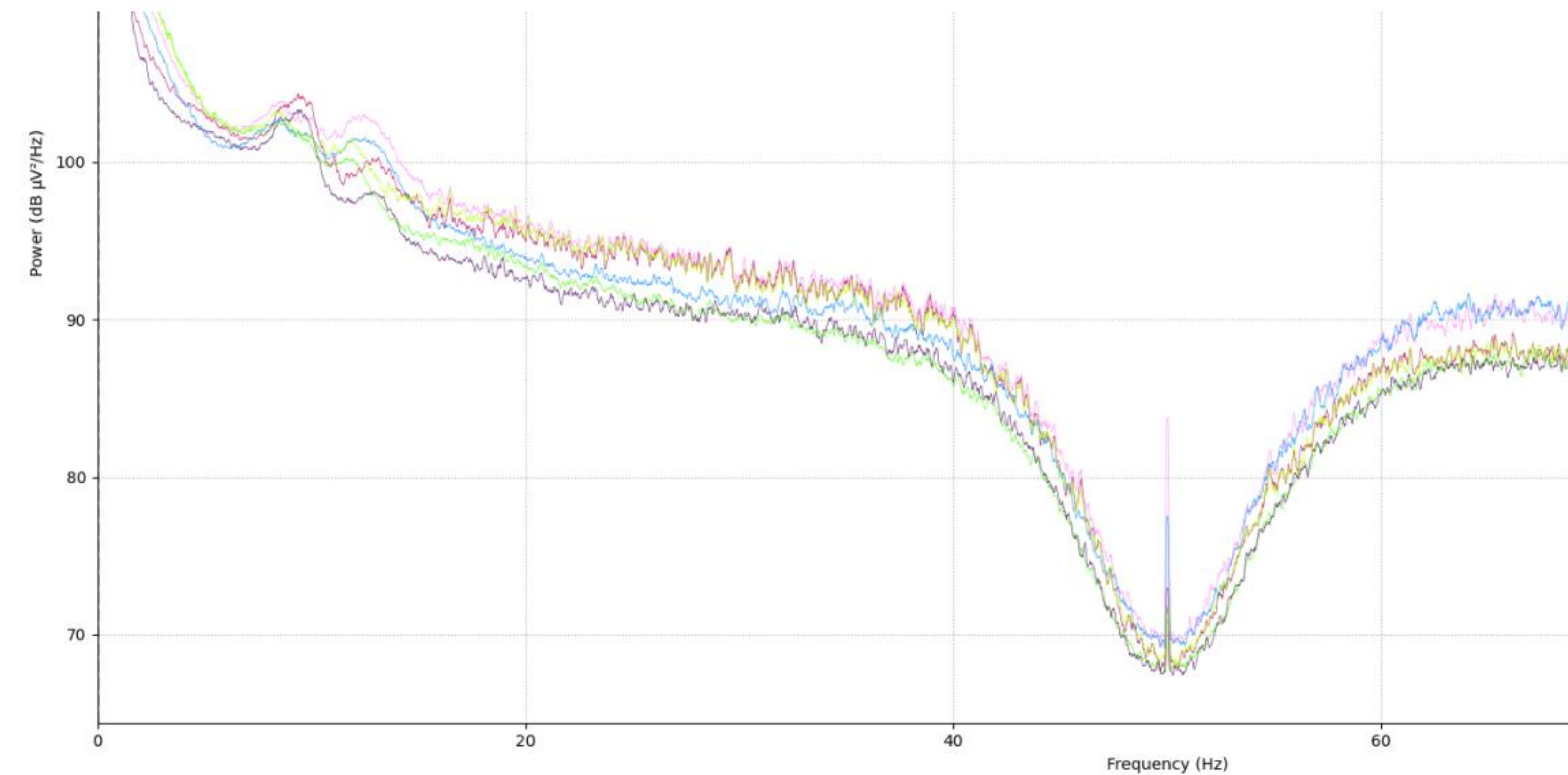
AHI predictor:

- “ODI”, “weight”, “BPdia”

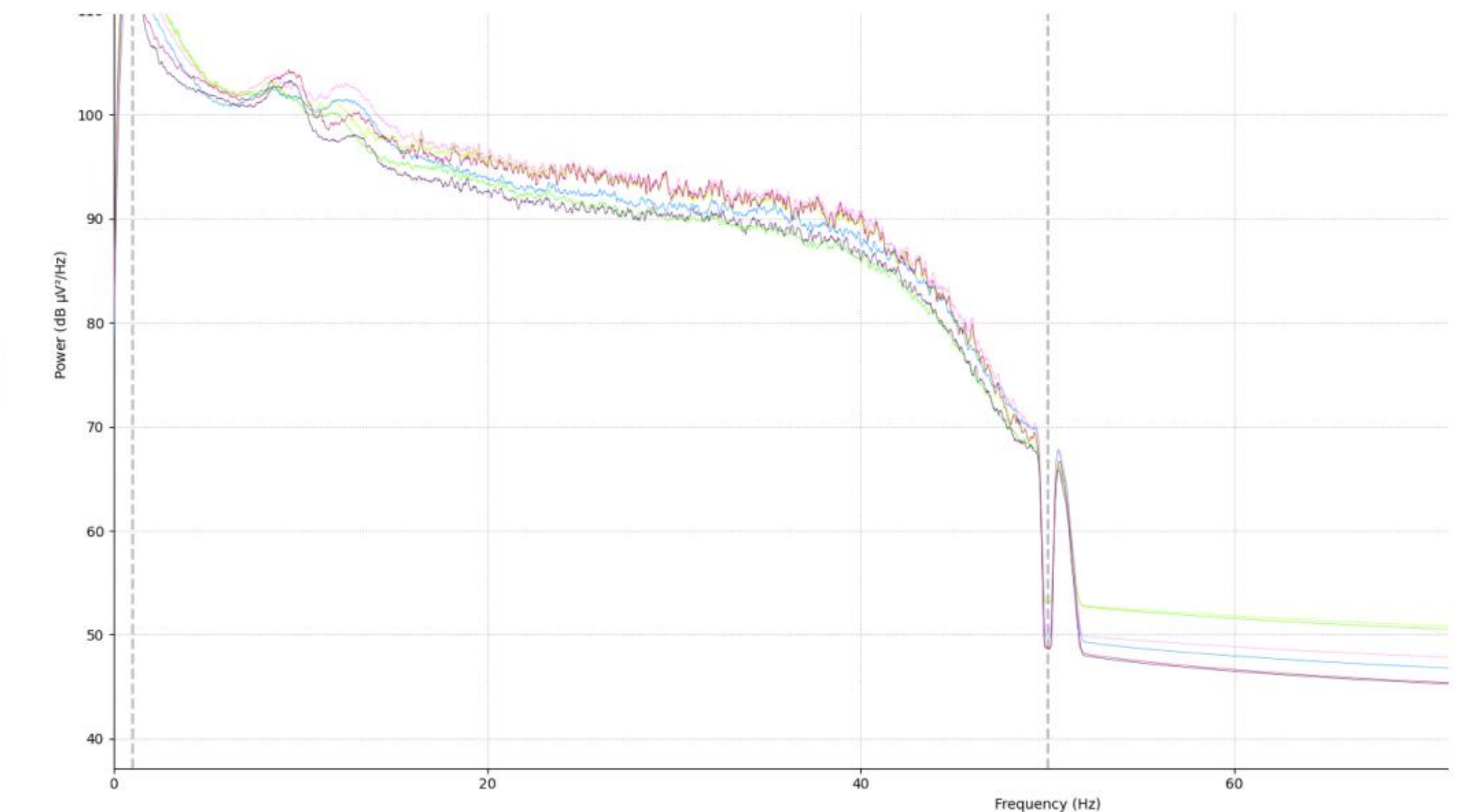
3. EEG Data: Preprocessing, Embeddings

EEG Preprocessing

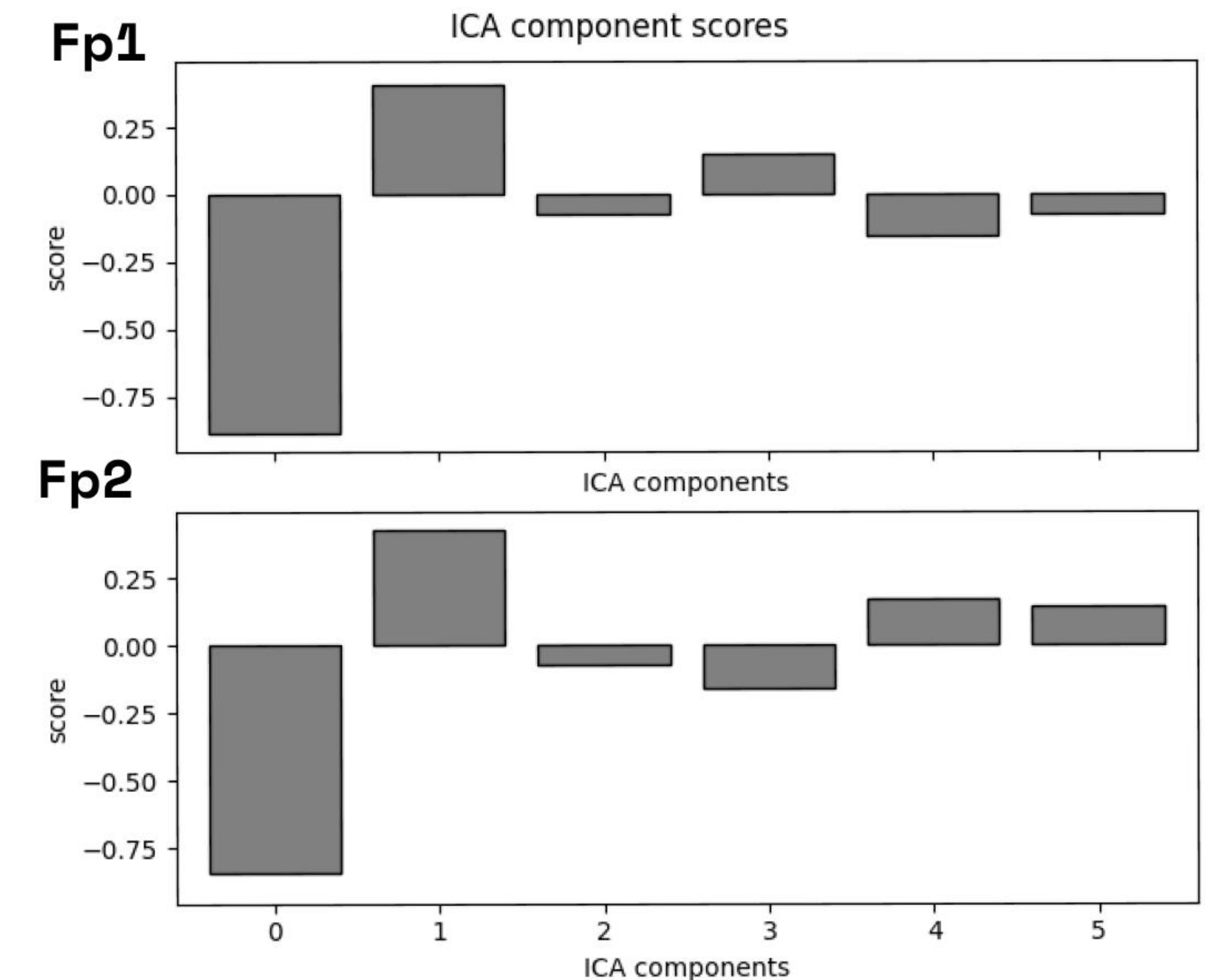
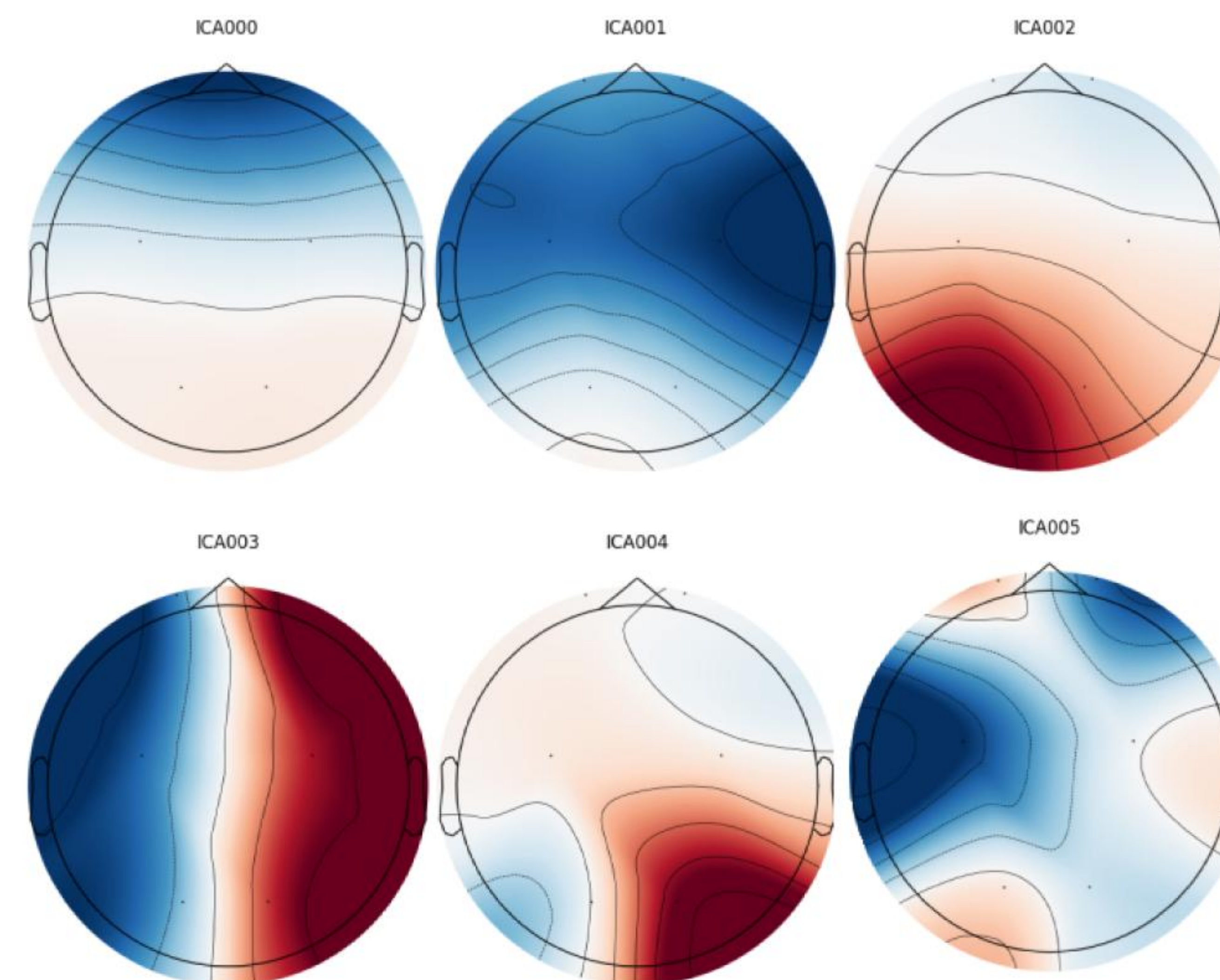
- **Notch filter (50 hz): removes power line noise**
- **Pass-band filter [1hz-50hz]: removes baseline drift, sweating, and electrode movement artifacts**
- **ICA: Computational and statistical technique used to separate a multivariate signal into additive, sub-components**
 - Eye movement interference
 - Heartbeat interference
 - Muscular interference (less relevant for sleep study)



Power Spectral Density of Patient 1 (Healthy), Night 1, prior to filtering operations.



Power Spectral Density of Patient 1 (Healthy), Night 1, after filtering operations.



The Learning Problem

- **Variable EEG Recording length:** how do we address this when creating embeddings?
 - *Crop to a fixed length (c.a. 7 hours)*
 - *Use a model which can handle this (usually deep-learning models)*
- **Input:** 8-hour EEG recording partitioned into windows
- **Labels:** Single global label per night (AHI score, apnea severity)
 - **Problem:** No per-window ground truth – only aggregate night-level annotations

(A) solution: Extract learned global embeddings from all windows in the night, then attempt to impute at the window level with models which allow it

- Many embeddings possibilities are available:
 - FPCA (Purely statistical, not standard)
 - Physiologically motivated (wavelets, power spectrum)
- ★ → Hybrid (per-window statistics, physiologically motivated)

EEGs & Biometric Data: It's Not Trivial

PROBLEM:

- EEG Is High-Dimensional And Noisy
- Predictive Features Require Strong Domain Assumptions (Bands, Channels, Windows)
- Poor Features Add Noise Rather Than Signal

Flexibility Vs. Robustness Tradeoff

- Fusing Tabular + Time Series Data Requires More Flexible Models
- Flexibility Comes At The Cost Of Robustness With Small Samples

Data-Hungry Algorithms With Few Samples

- Neural Networks And Attention-Based Models Need Large Datasets: With ~78 Samples Generalization Is Very Optimistic

Data leakage and embeddings

What Is Data Leakage

- Leakage Occurs When Information From The Test Set Influences The Training Pipeline

The Problem With FPCA And Wavelets

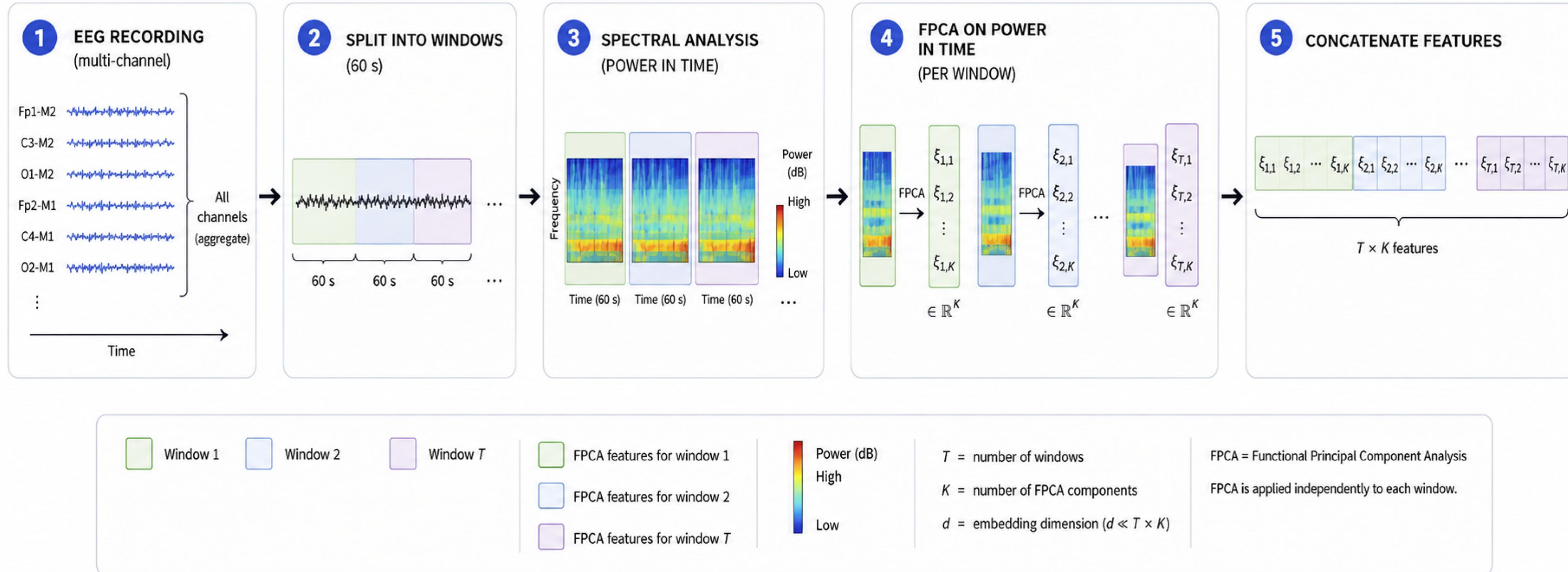
- Both Methods Require Fitting A Transformation Before Projecting Data
- FPCA: The Functional Basis And Principal Components Are Estimated From The Data
- Wavelets: If Adaptive (E.G. Learned Filterbanks) Or If Statistics Are Used For Normalization, The Same Issue Applies

→ If Fitted On The Full Dataset Before Splitting, Test Samples Have Already Influenced The Embedding Space

Correct Procedure

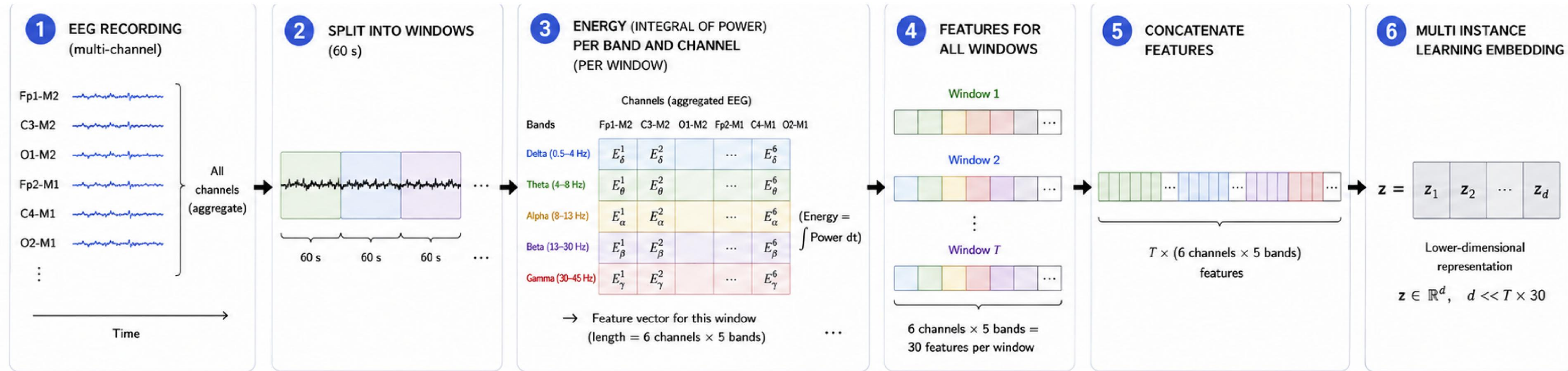
- Fit FPCA / Compute Normalization Statistics On The Training Set Only
- Apply The Learned Transformation To The Test Set Without Refitting
- Identical To How StandardScaler Or PCA Should Be Used Inside A Sklearn Pipeline

Power FPCA



- EEG Epochs Are Transformed Into Spectral Band Powers (Delta, Theta, Alpha And Beta)
- For Each Channel And Frequency Band, The Temporal Evolution Of Power Across The Recording Is Modeled As A Functional Curve And Analyzed Using FPCA.

Raw Power

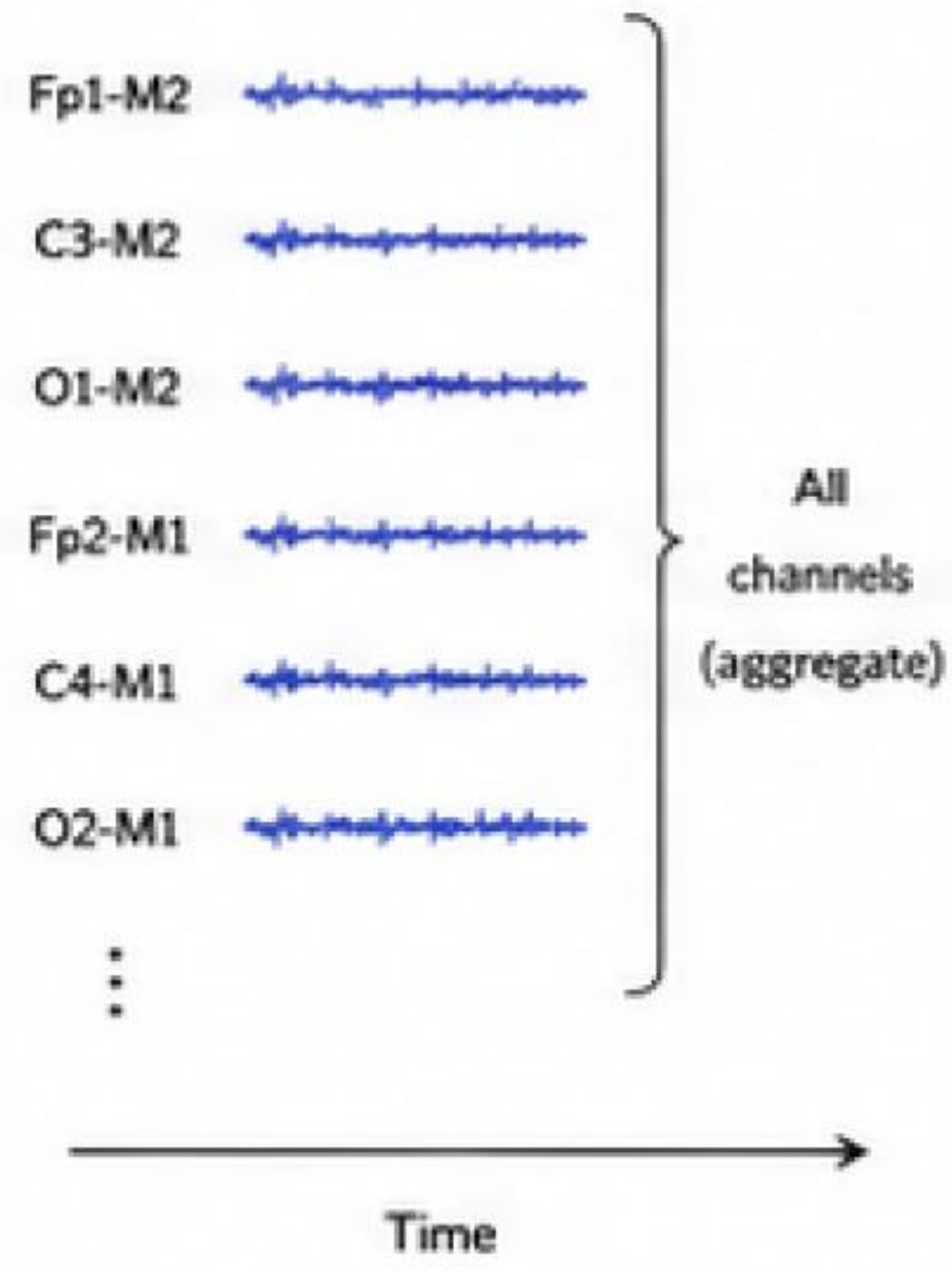


Window 1	Delta (0.5–4 Hz)	E_b^c : Energy (integral of power) for channel c (1...6) in band b for the current window	T = number of windows (each window = 60 s)	Feature per window = $[E_{\delta}^1, \dots, E_{\delta}^6, E_{\theta}^1, \dots, E_{\theta}^6, E_{\alpha}^1, \dots, E_{\alpha}^6, E_{\beta}^1, \dots, E_{\beta}^6, E_{\gamma}^1, \dots, E_{\gamma}^6]$ (concatenation of energies for all channels across all bands)
Window 2	Theta (4–8 Hz)		6 = number of channels (aggregated EEG)	Final concatenated feature vector length = $T \times 30$
Window T	Alpha (8–13 Hz)	$c \in \{1, \dots, 6\}$ channels	5 = number of frequency bands	MIL embedding dimension: $d \ll T \times 30$
	Beta (13–30 Hz)	$b \in \{\delta, \theta, \alpha, \beta, \gamma\}$ bands	30 = 6×5 features per window	
	Gamma (30–45 Hz)			

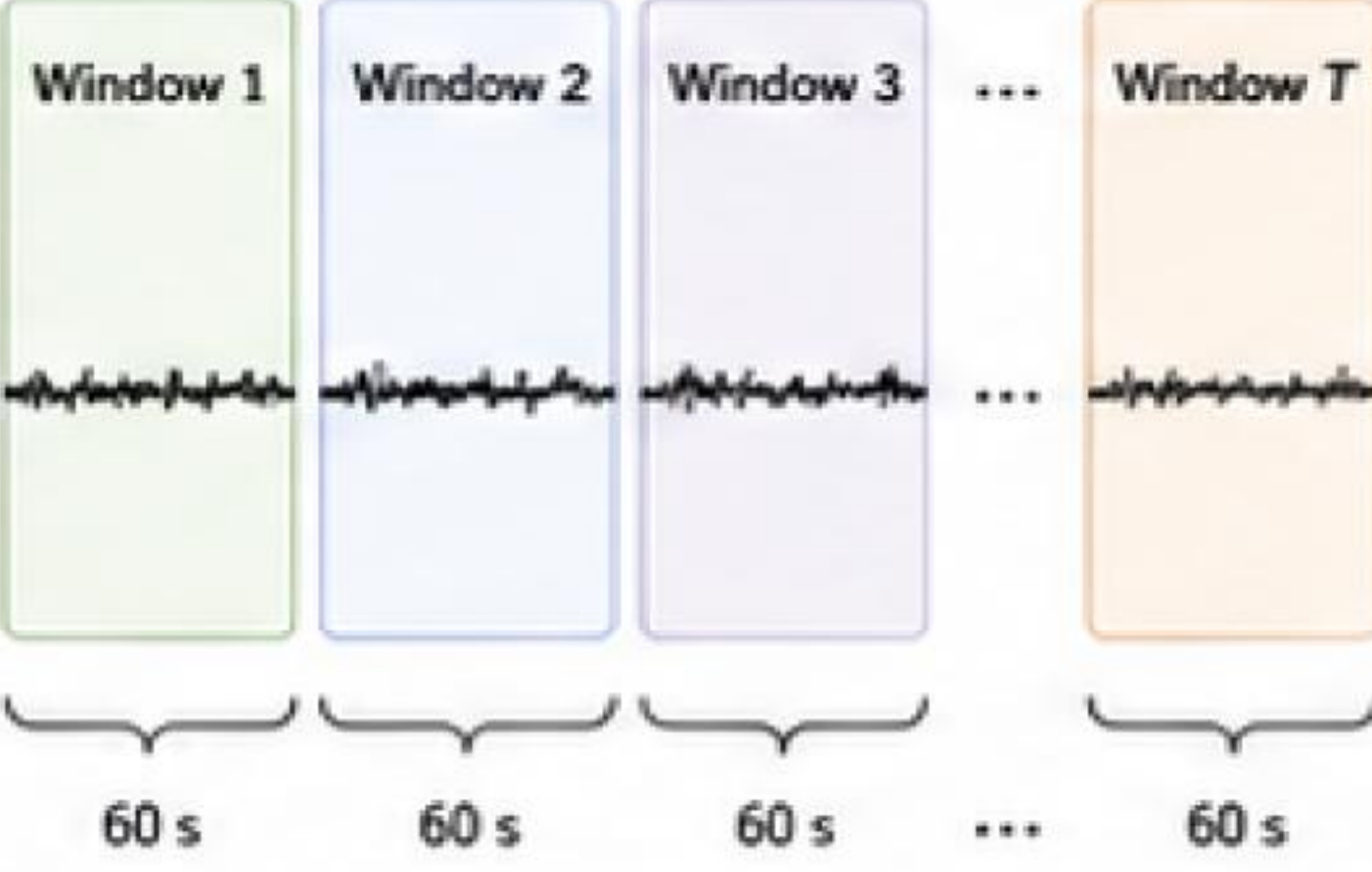
- The Power Spectral Density (PSD) Is Computed For Each EEG Epoch And Channel
- Power Values Are Integrated Within Physiologically Relevant Frequency Bands (Delta, Theta, Alpha, Beta And Gamma) To Obtain Energy (Integral Of Power)

Wavelets

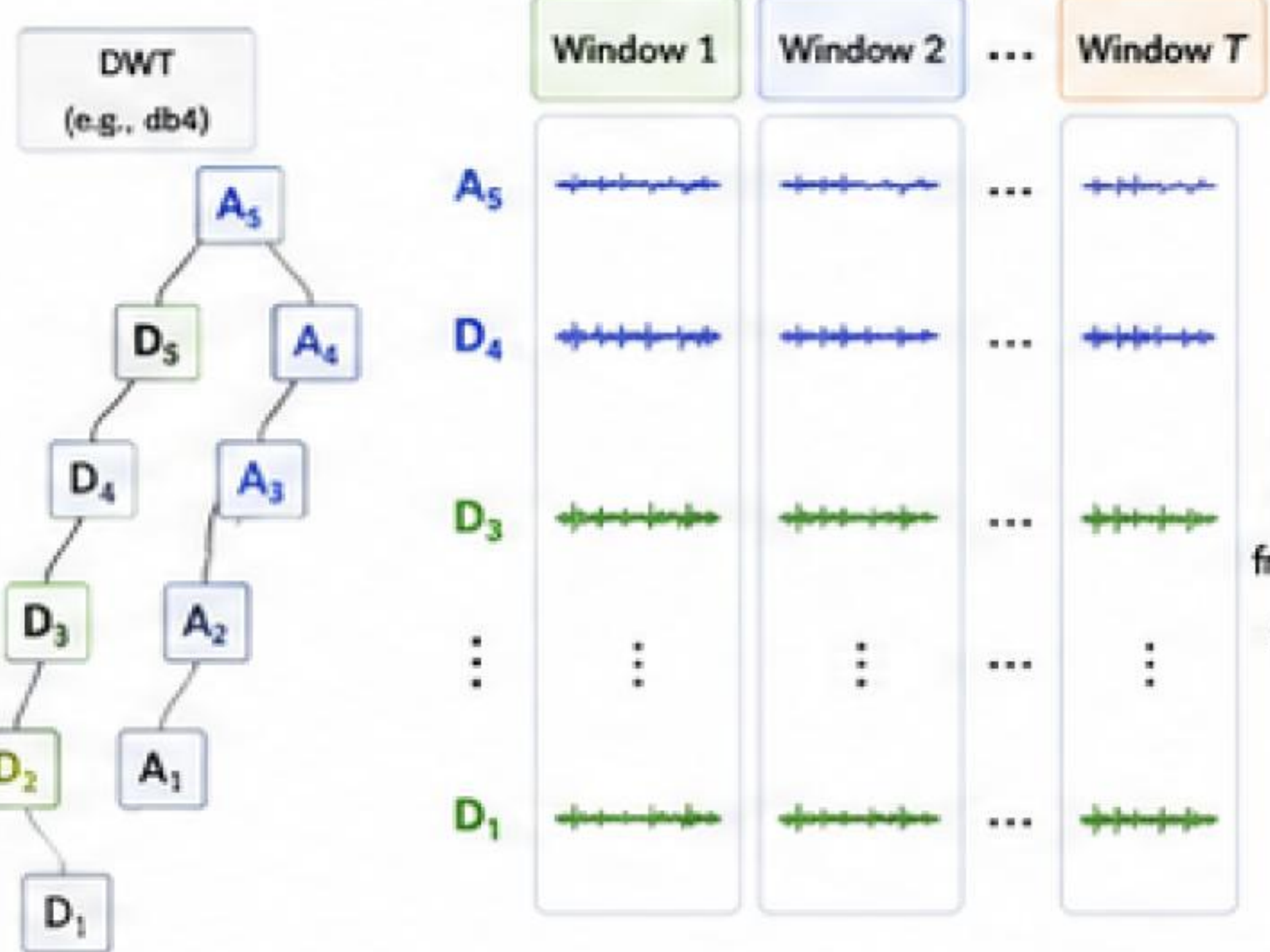
1 EEG RECORDING (multi-channel)



2 SPLIT INTO WINDOWS (60 s epochs)



3 WAVELET DECOMPOSITION (per window)



4 FEATURES PER SUBBAND (per window)

Subband	Metrics (per window)		
	Power	Mean	Std
A_5			
D_4			
D_3			
$\vdots$			

Window 1	Window 2	...	Window T
$x_{A_5}^{(1)} \in \mathbb{R}^3$	$x_{A_5}^{(2)} \in \mathbb{R}^3$	...	$x_{A_5}^{(T)} \in \mathbb{R}^3$
$x_{D_4}^{(1)} \in \mathbb{R}^3$	$x_{D_4}^{(2)} \in \mathbb{R}^3$	...	$x_{D_4}^{(T)} \in \mathbb{R}^3$
$x_{D_3}^{(1)} \in \mathbb{R}^3$	$x_{D_3}^{(2)} \in \mathbb{R}^3$	...	$x_{D_3}^{(T)} \in \mathbb{R}^3$
$\vdots$	$\vdots$		$\vdots$

5 AGGREGATE OVER THE NIGHT (across windows)

Subband	Aggregation metrics (across all windows)	
	Mean	Std
A_5	μ_{A_5}	σ_{A_5}
D_4	μ_{D_4}	σ_{D_4}
D_3	μ_{D_3}	σ_{D_3}
$\vdots$		

For each subband, compute mean and std of each feature across all windows.

→ $2 \times 3 = 6$ values per subband

5 MEHTT (per NIGHT VECTOR)

D_3						D_4						A_5					
$\mu_{D_3, Power}$	$\sigma_{D_3, Power}$	$\mu_{D_3, Mean}$	$\sigma_{D_3, Mean}$	$\mu_{D_3, Std}$	$\sigma_{D_3, Std}$	$\mu_{D_4, Power}$	$\sigma_{D_4, Power}$	$\mu_{D_4, Mean}$	$\sigma_{D_4, Mean}$	$\mu_{D_4, Std}$	$\sigma_{D_4, Std}$	$\mu_{A_5, Power}$	$\sigma_{A_5, Power}$	$\mu_{A_5, Mean}$	$\mu_{A_5, Std}$	$\sigma_{A_5, Std}$	

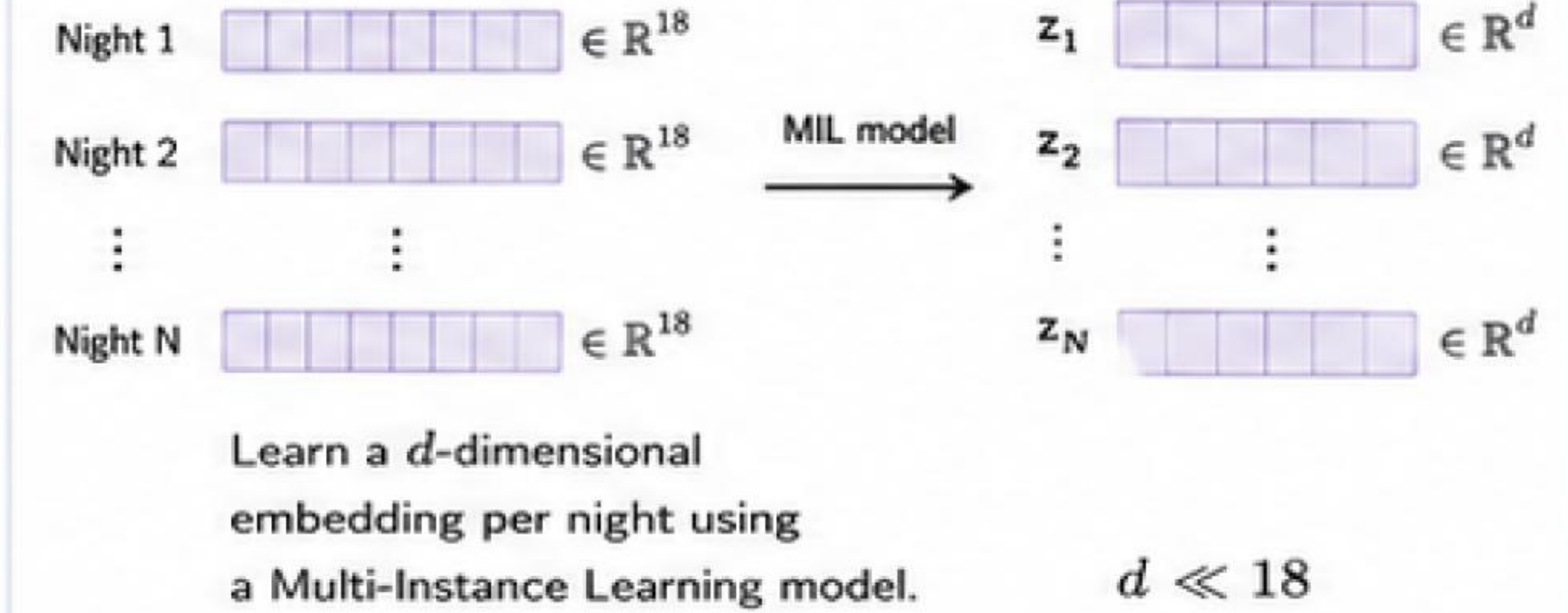
18-dimensional feature vector per night $\in \mathbb{R}^{18}$

6 NIGHT FEATURE VECTOR

D_3						D_4						A_5					
$\mu_{D_3, Power}$	$\sigma_{D_3, Power}$	$\mu_{D_3, Mean}$	$\mu_{D_3, Std}$	$\sigma_{D_3, Std}$		$\mu_{D_4, Power}$	$\sigma_{D_4, Power}$	$\mu_{D_4, Mean}$	$\mu_{D_4, Std}$	$\sigma_{D_4, Std}$		$\mu_{A_5, Power}$	$\sigma_{A_5, Power}$	$\mu_{A_5, Mean}$	$\mu_{A_5, Std}$	$\sigma_{A_5, Std}$	

18-dimensional feature vector per night $\in \mathbb{R}^{18}$

7 MULTI INSTANCE LEARNING EMBEDDING



- Window 1
- Window 2
- Window 3
- Window T
- $\vdots$

Delta (0.5–4 Hz)

Theta (4–8 Hz)

Alpha (8–13 Hz)

Beta (13–30 Hz)

Gamma (30–45 Hz)

Bands $b \in \{\delta, \theta, \alpha, \beta, \gamma\}$

DWT : Discrete Wavelet Transform

db4 : Daubechies wavelet (order 4)

A_j : Approximation at level j

D_j : Detail at level j

We keep subbands from D_3 to A_5 (inclusive).

Per window we compute for each selected subband and each channel:

- Power (integral of power)
- Mean
- Std (standard deviation)

→ 3 metrics per subband.

T = number of windows (epochs) (each window = 60 s)

6 = number of channels (aggregated EEG)

5 = number of subbands used (D_3, D_4, A_5)

Feature per window = 5 subbands $\times$ 3 metrics $\times$ 6 channels = 90

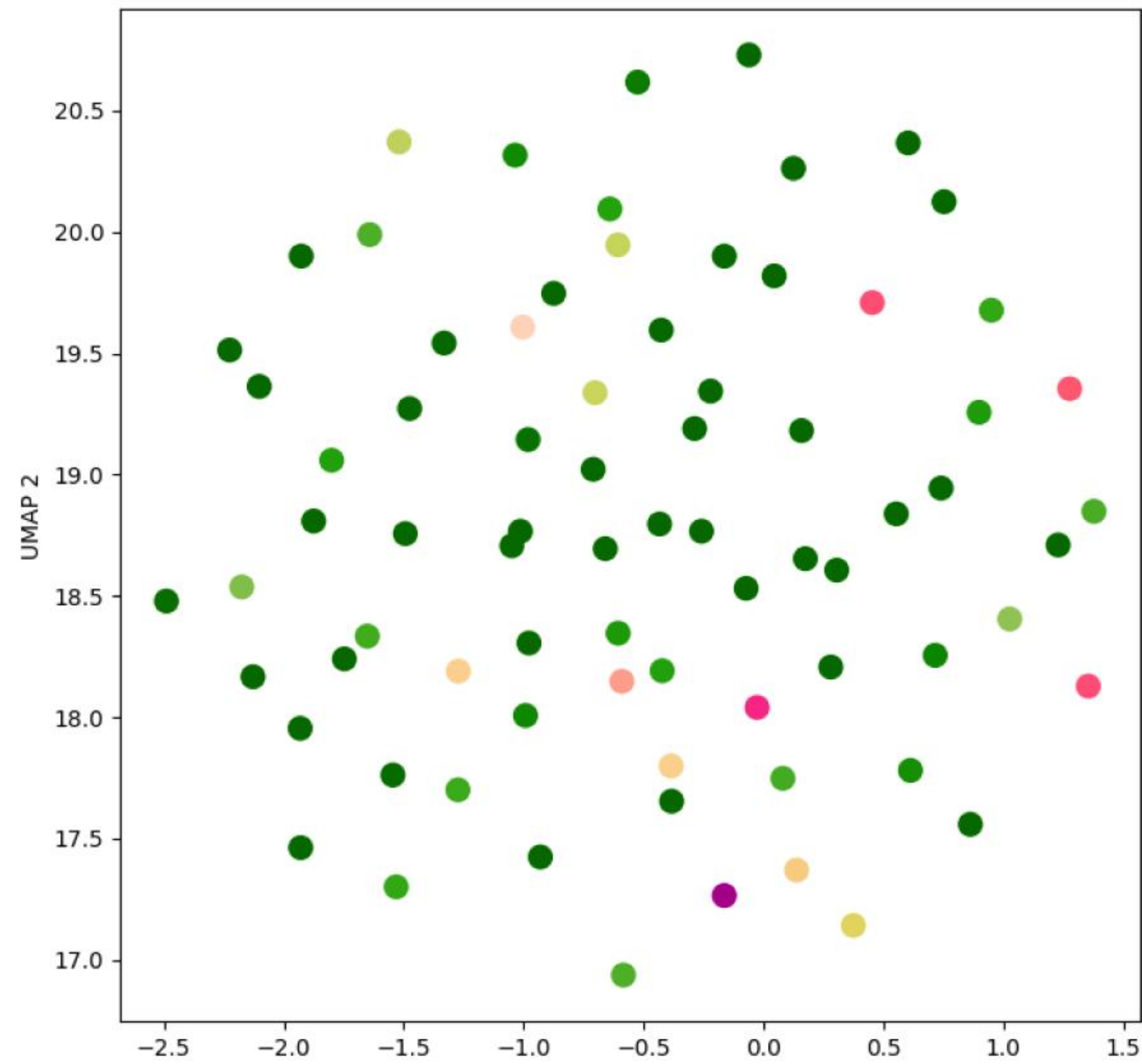
Night feature vector = mean & std across windows for each (subband, metric, channel):

$5 \times 3 \times 6 \times 2 = 180 \rightarrow$ grouped as 18 values ((5 $\times$ 3) metrics per channel aggregated as mean & std) = 18

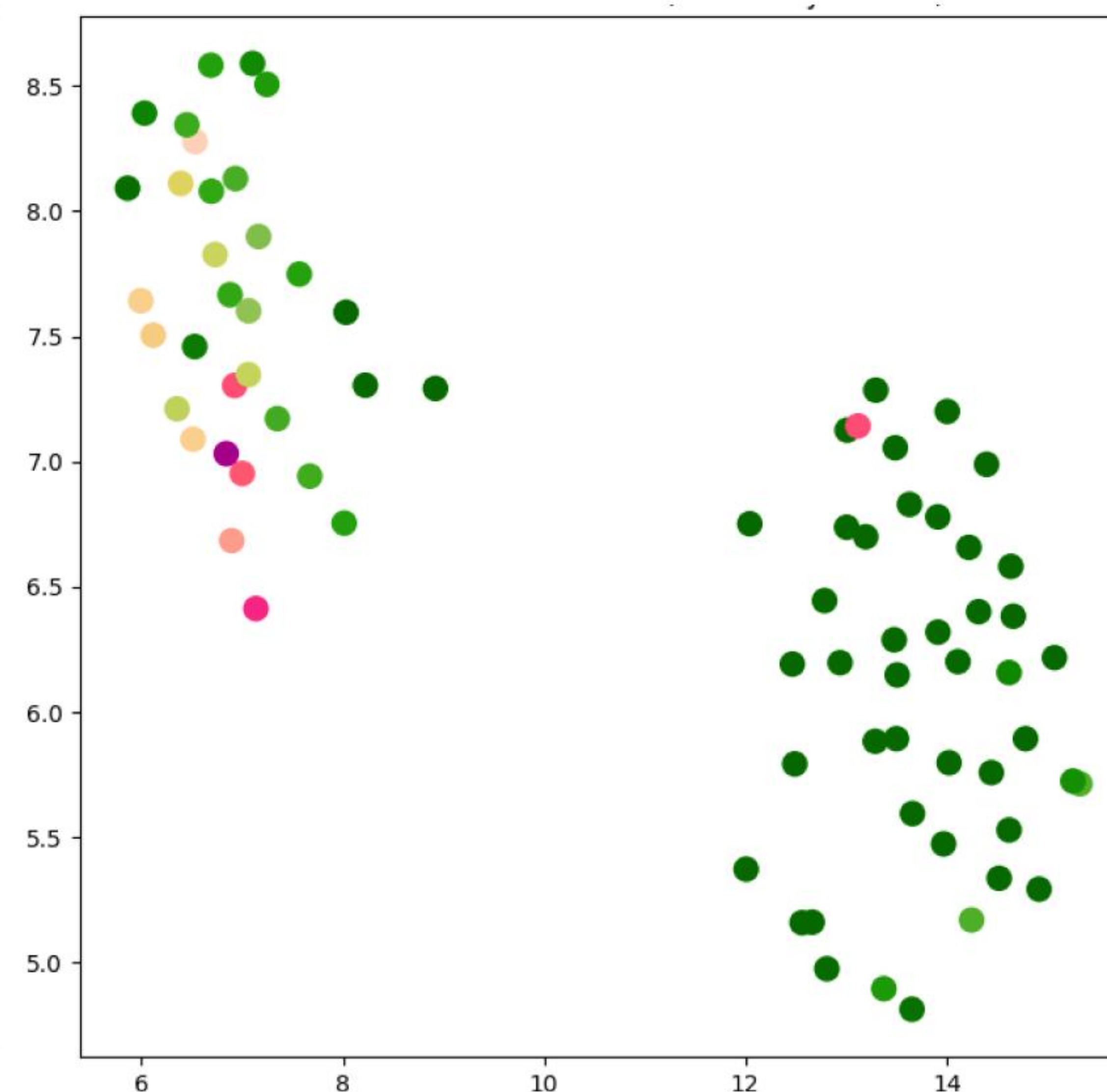
MIL embedding dimension: $d \ll 18$

Evaluating Embedding Expressiveness (UMAP & PCA)

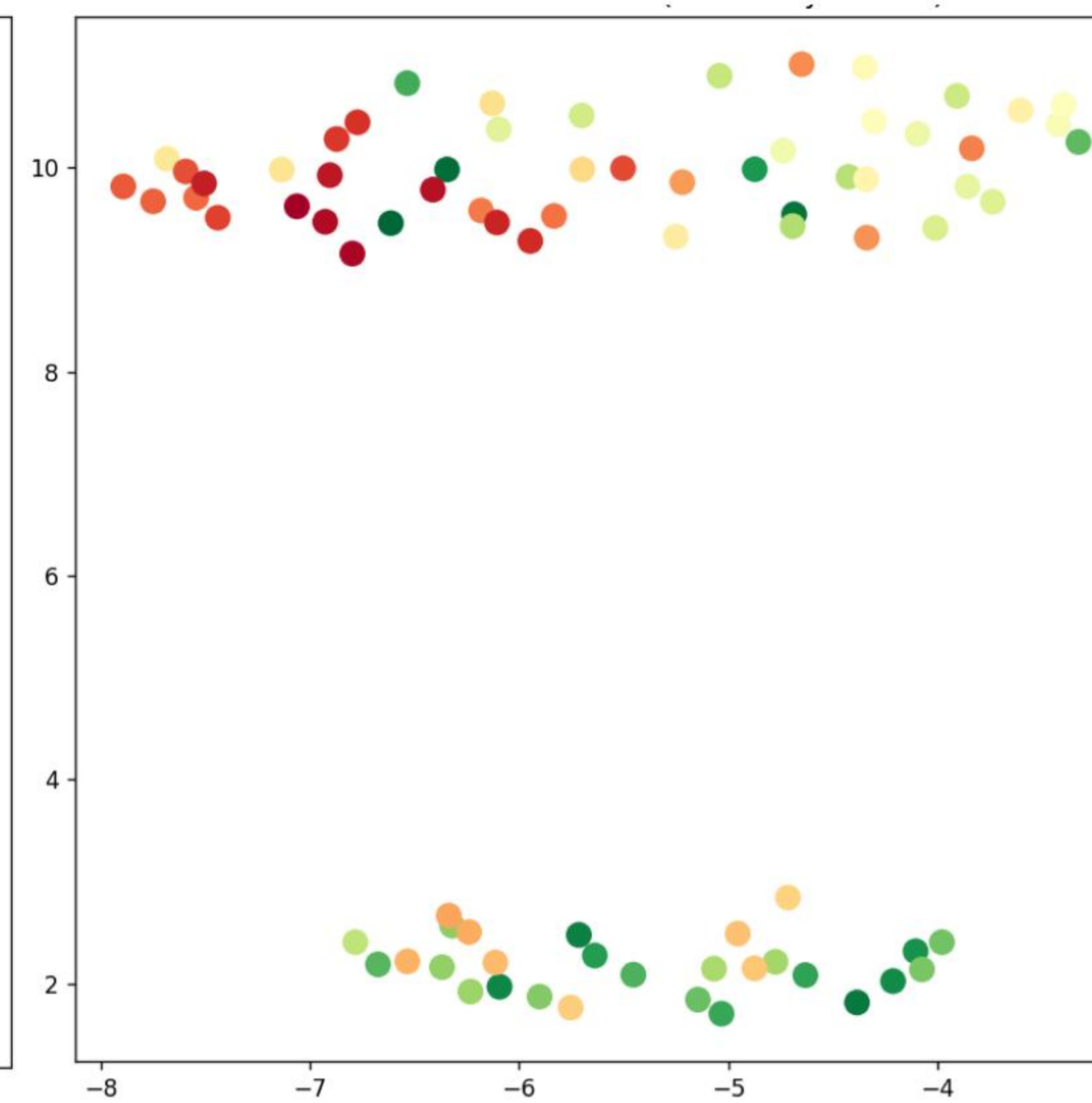
FPCA



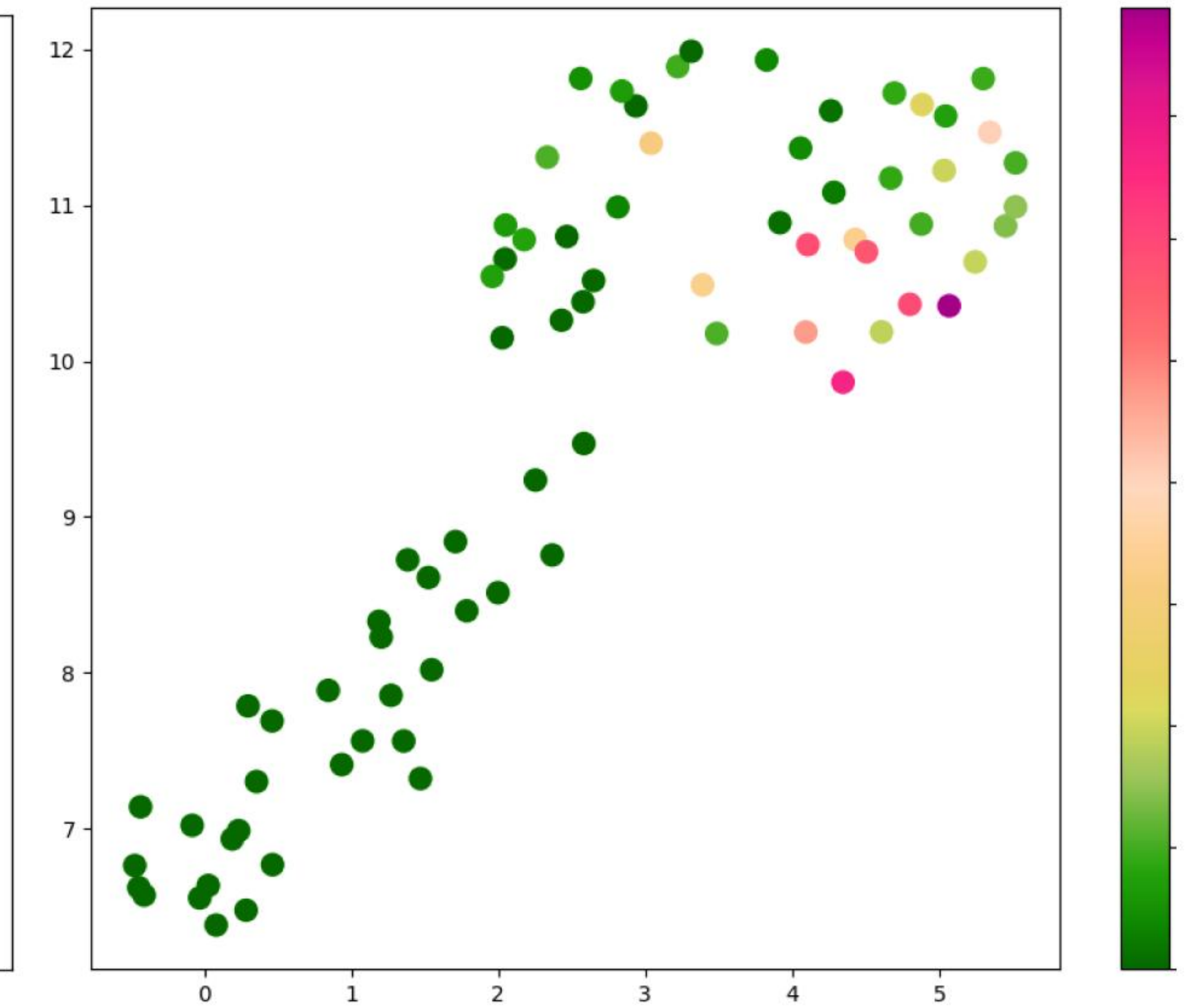
Power FPCA



Wavelet



Raw Power



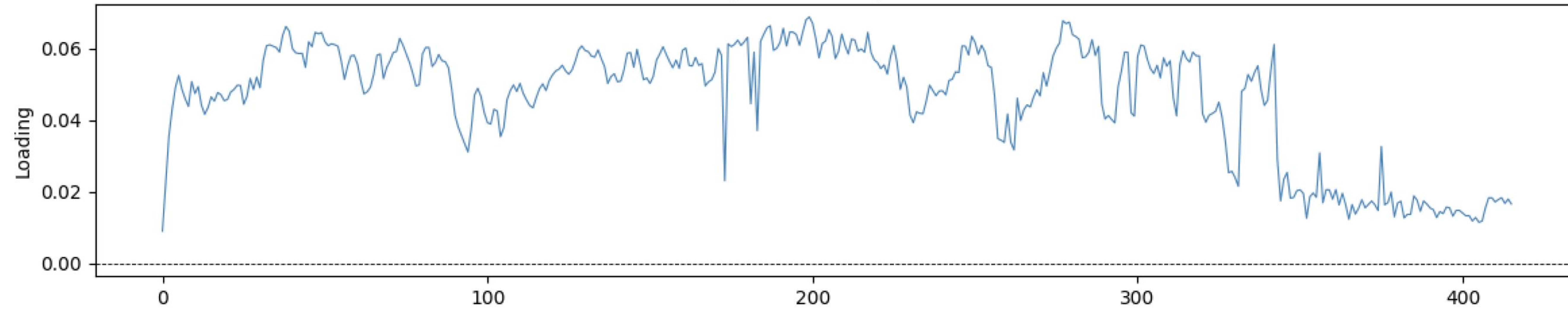
2d-UMAP reduction to visualize semantic meaningfulness of the embeddings

We can gain some insight into how meaningful the embeddings are for our end task by extracting PC and using UMAP against our class division

→ We see that wavelet and raw power embeddings appear more promising, with FPCA being completely dominated by other phenomena

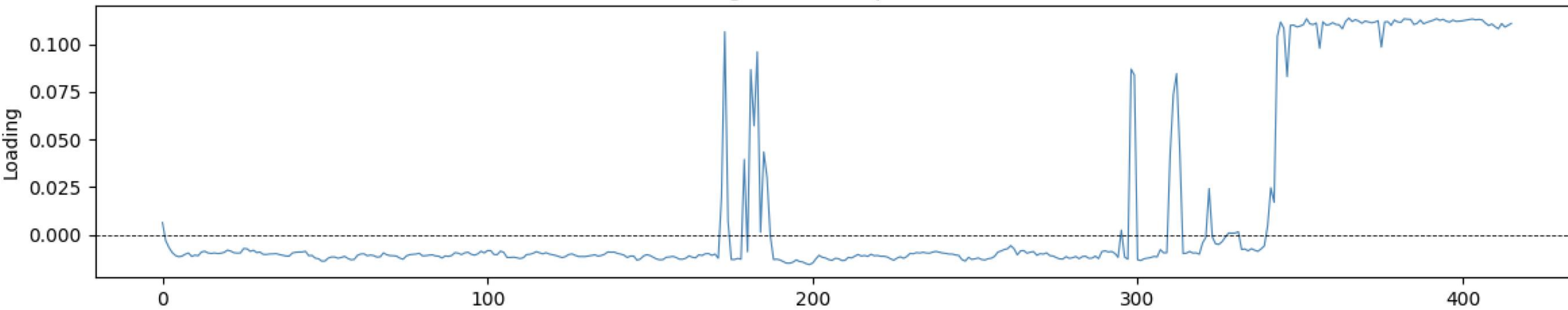
PCA Loadings

PC1 Loadings (35.9% explained variance)



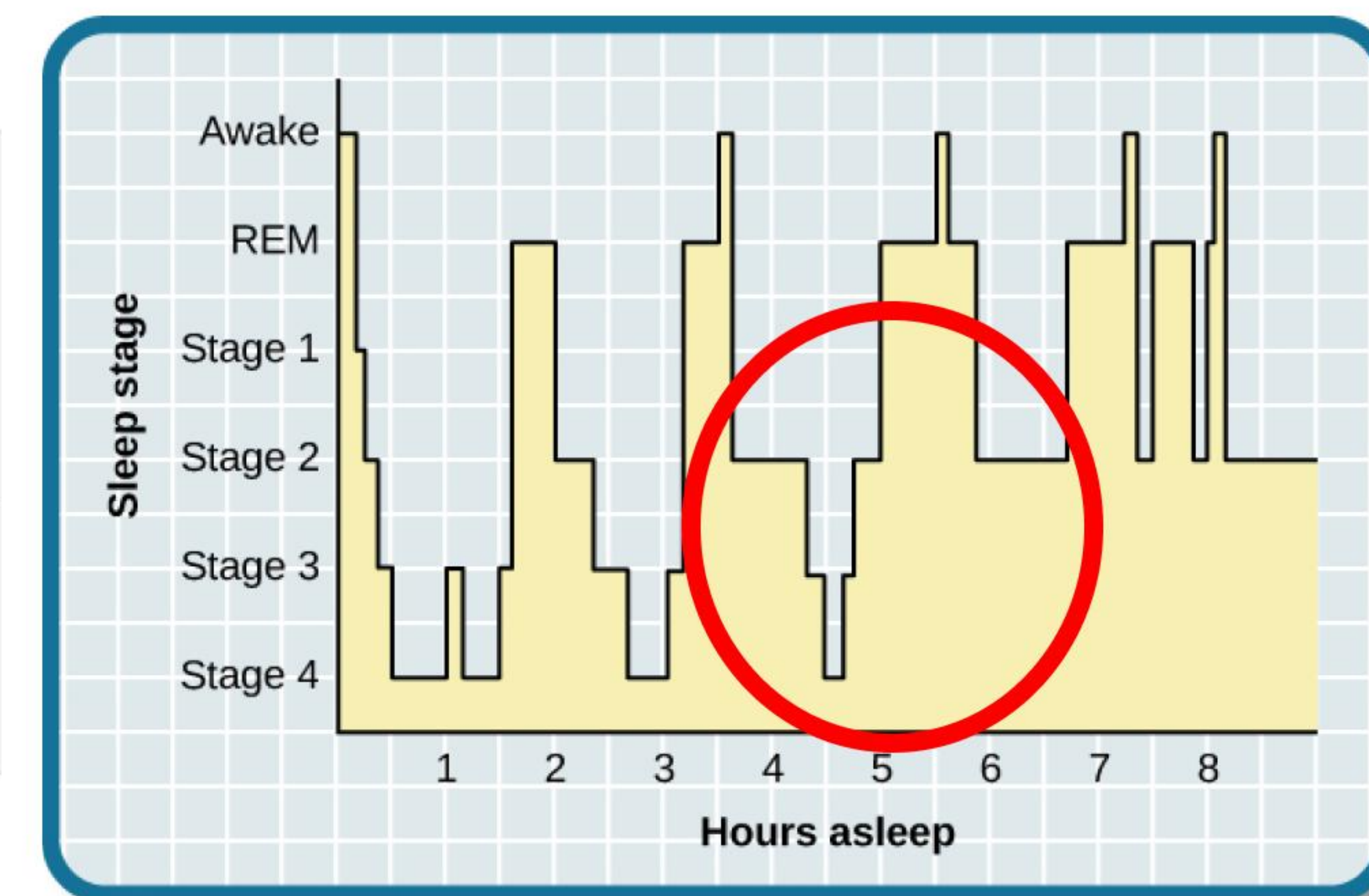
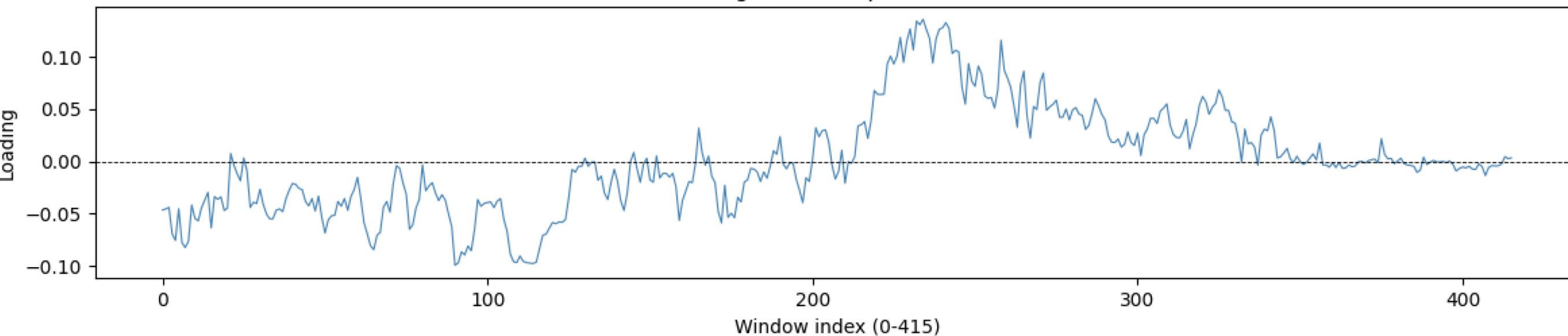
sleep onset &
heavy sleep

PC2 Loadings (18.1% explained variance)

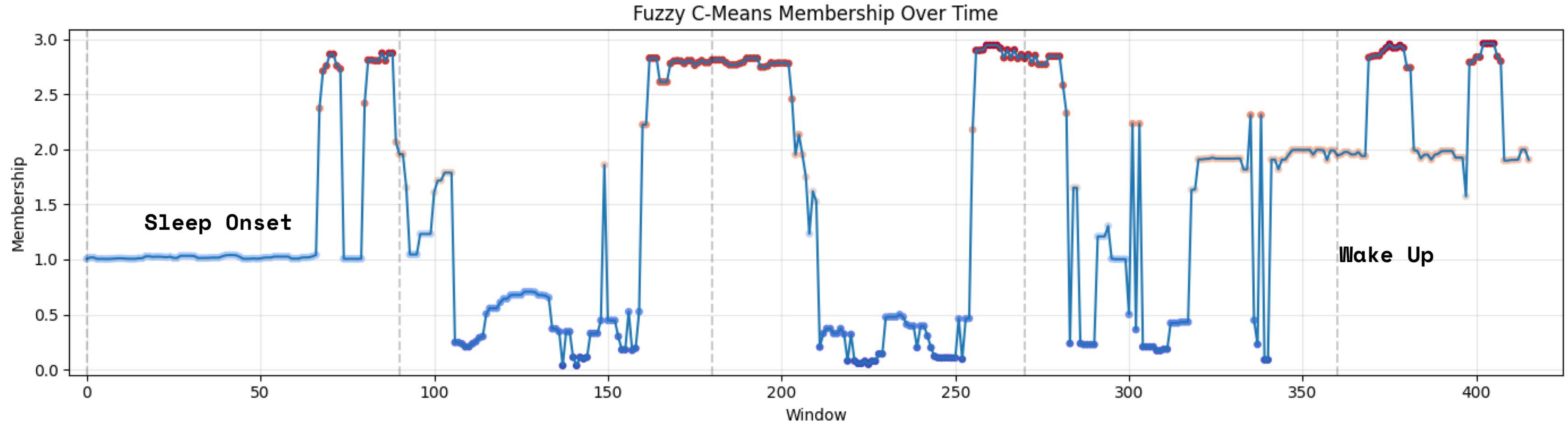


wake-up
period

PC3 Loadings (6.2% explained variance)



Periodicity (C-Means)



C-Means Objective

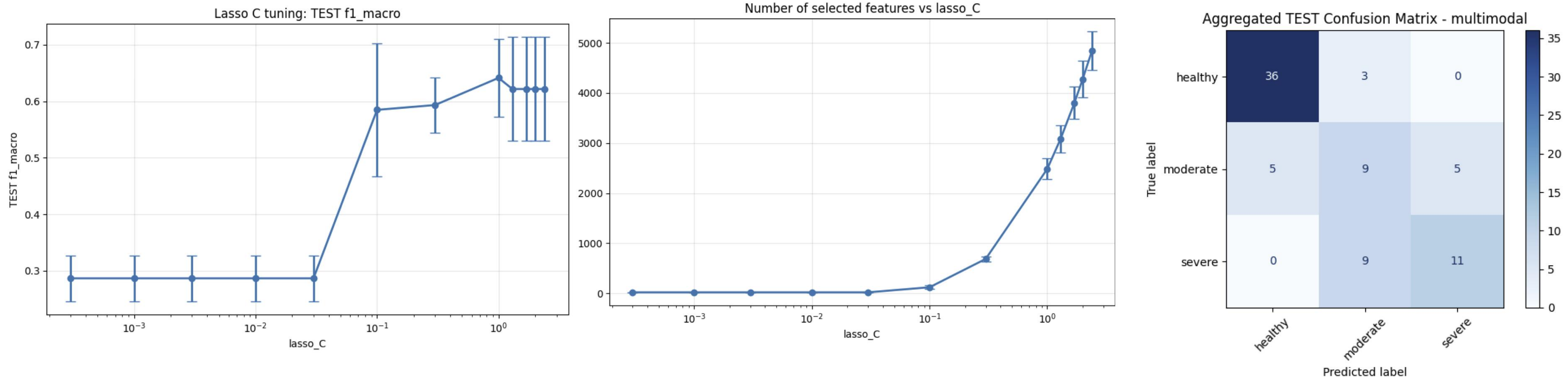
$$J(W, C) = \sum_{i=1}^n \sum_{j=1}^c w_{ij}^m \|\mathbf{x}_i - \mathbf{c}_j\|^2$$

We compute the weighted average of the cluster index to visualize smooth changes in cluster indices

Problem: EEG data encodes a large amount of phenomenon which reflect the extreme complexity of brain activity, so it is not trivial to see how these embeddings can be used effectively.

4. Linear Models for Classification & Regression

Multi-Modal Logistic Classifier

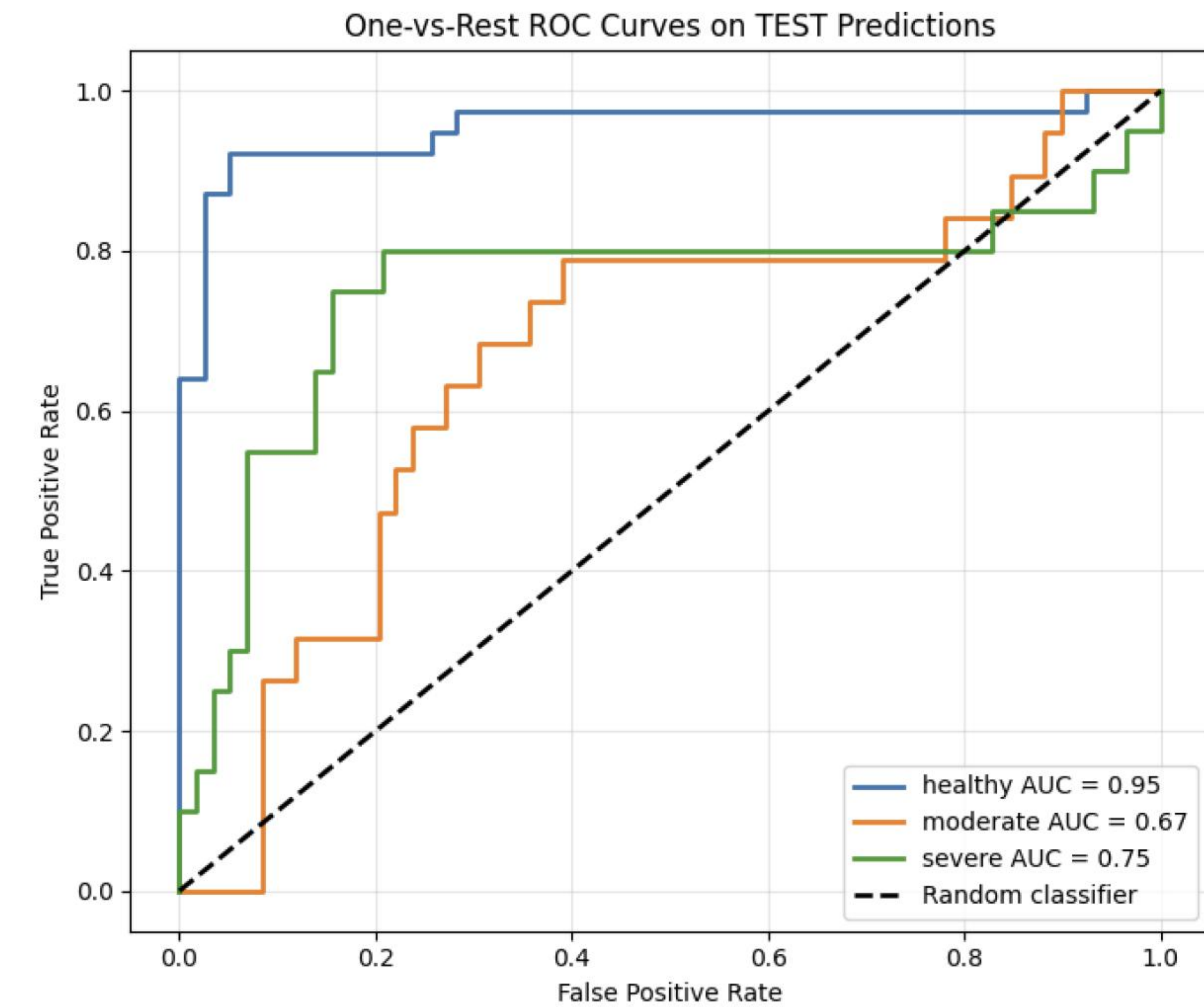
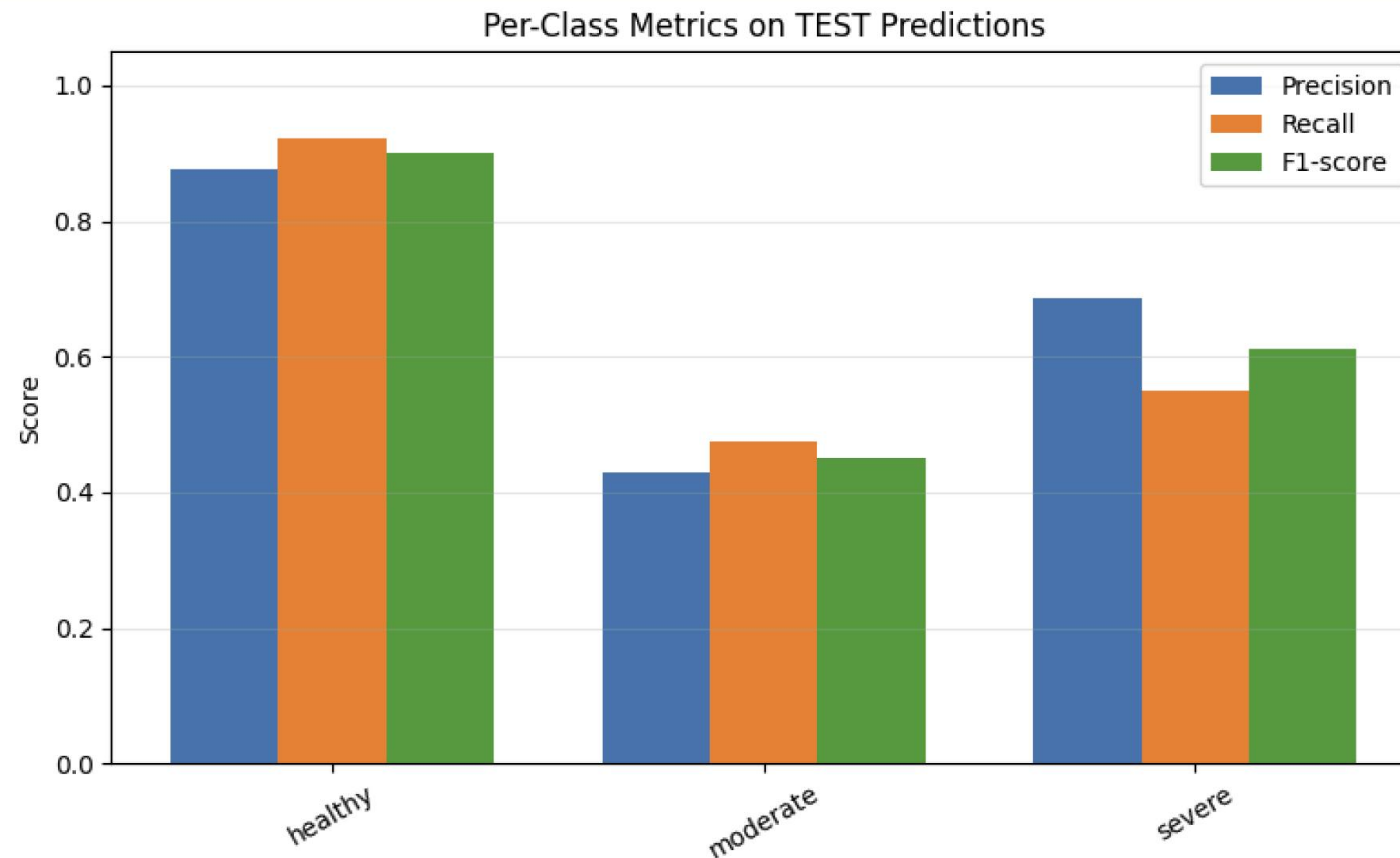


We use a multi-modal logistic classifier with a L1 regularization, in order to obtain the convergence of the method.

Multi-modal: we are integrating together the EEG embeddings and the physiological data by simple concatenation

NOTE: As reported above, we have merged the “Mild” and “Moderate” categories together, so this is a 3-class classification problem

Multi-Modal Logistic Classifier (Metrics)

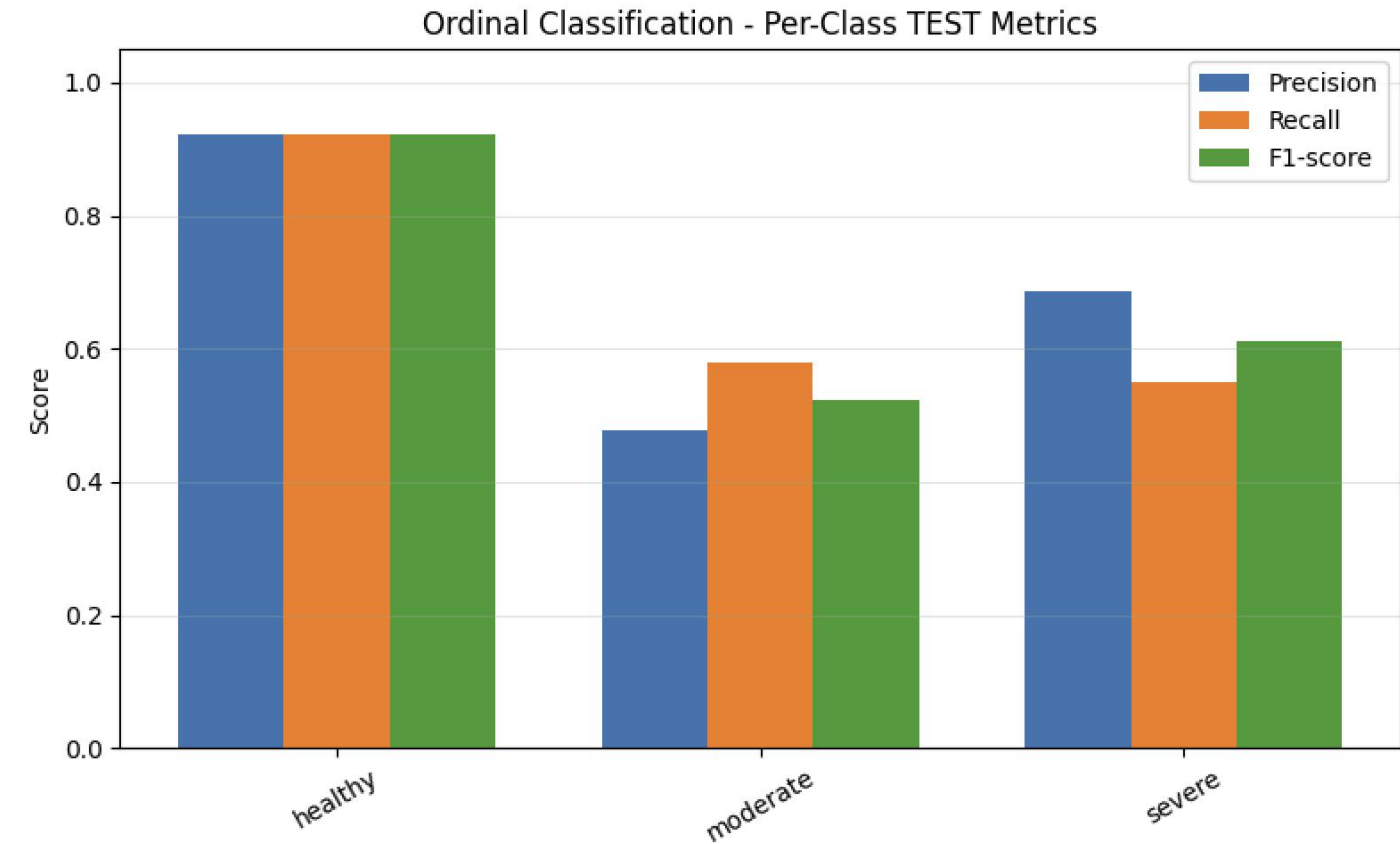
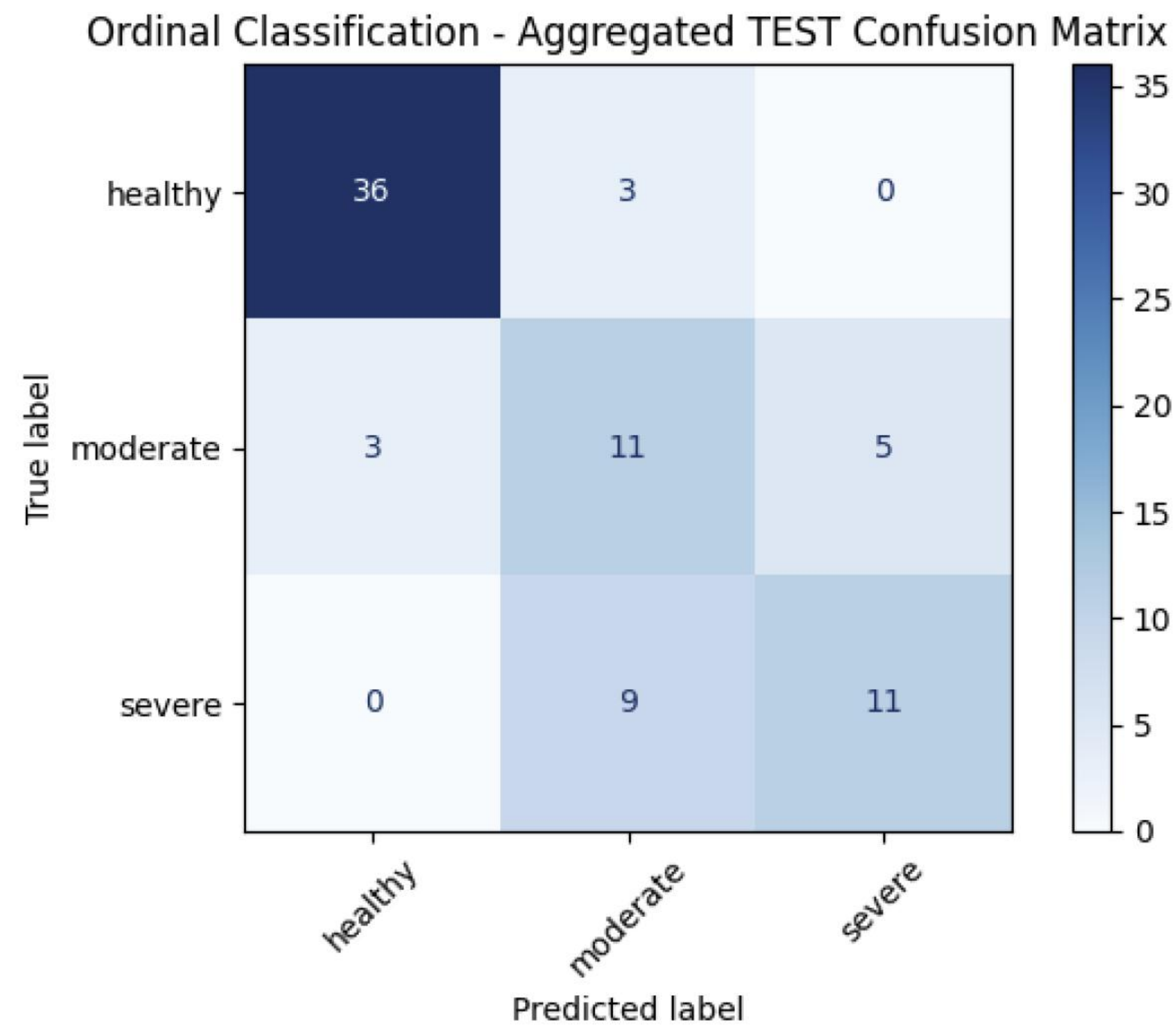


- The healthy class is the easiest to classify.
- The moderate class has the weakest performance, probably because it overlaps heavily with both healthy and severe cases → classifying extreme phenomena is easier.
- The ROC curves confirm this pattern: healthy has the highest AUC, while moderate and severe are harder to separate.

How can we address this?

→ Convert the problem into two binary splits

Multi-Modal Logistic Classifier



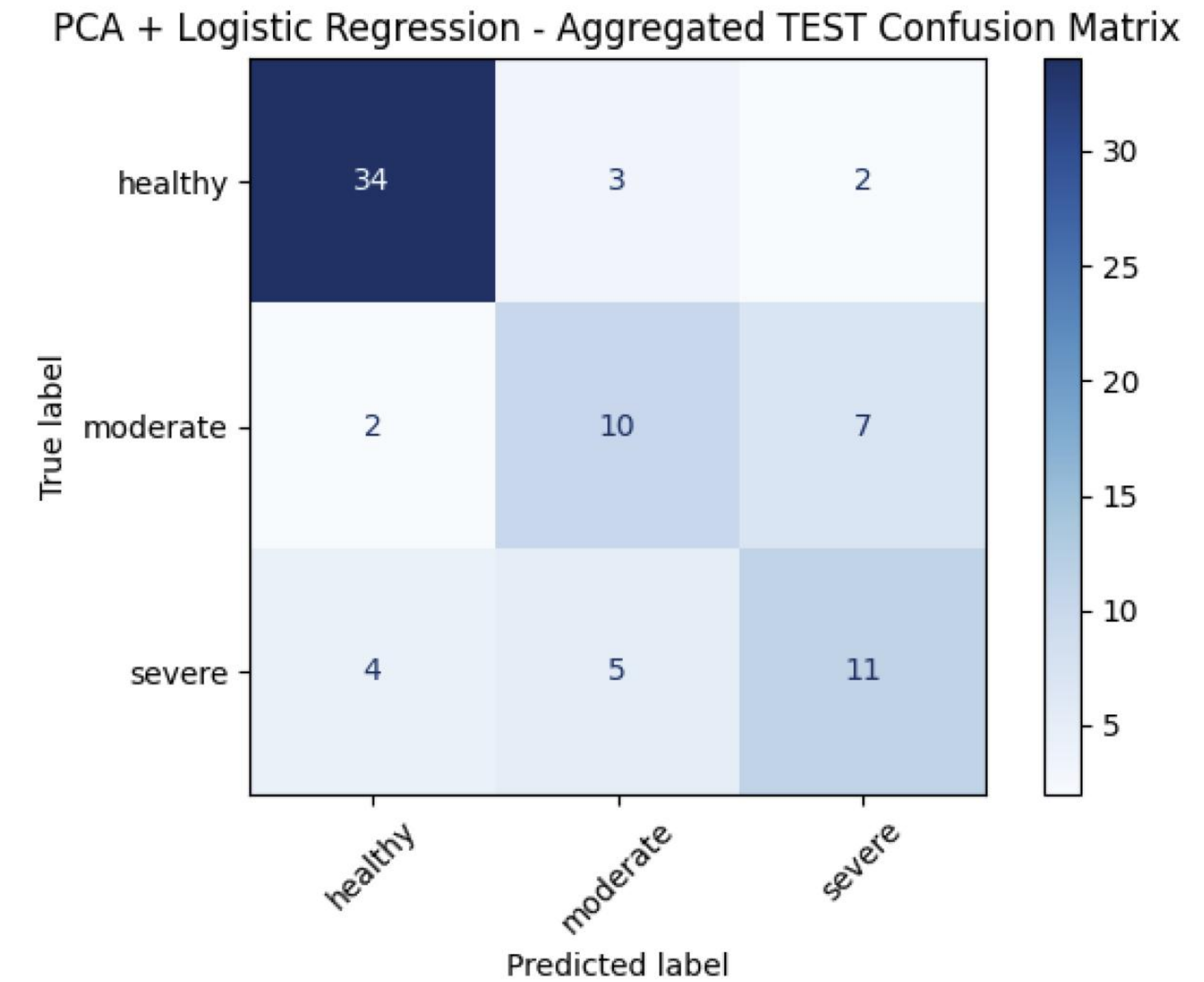
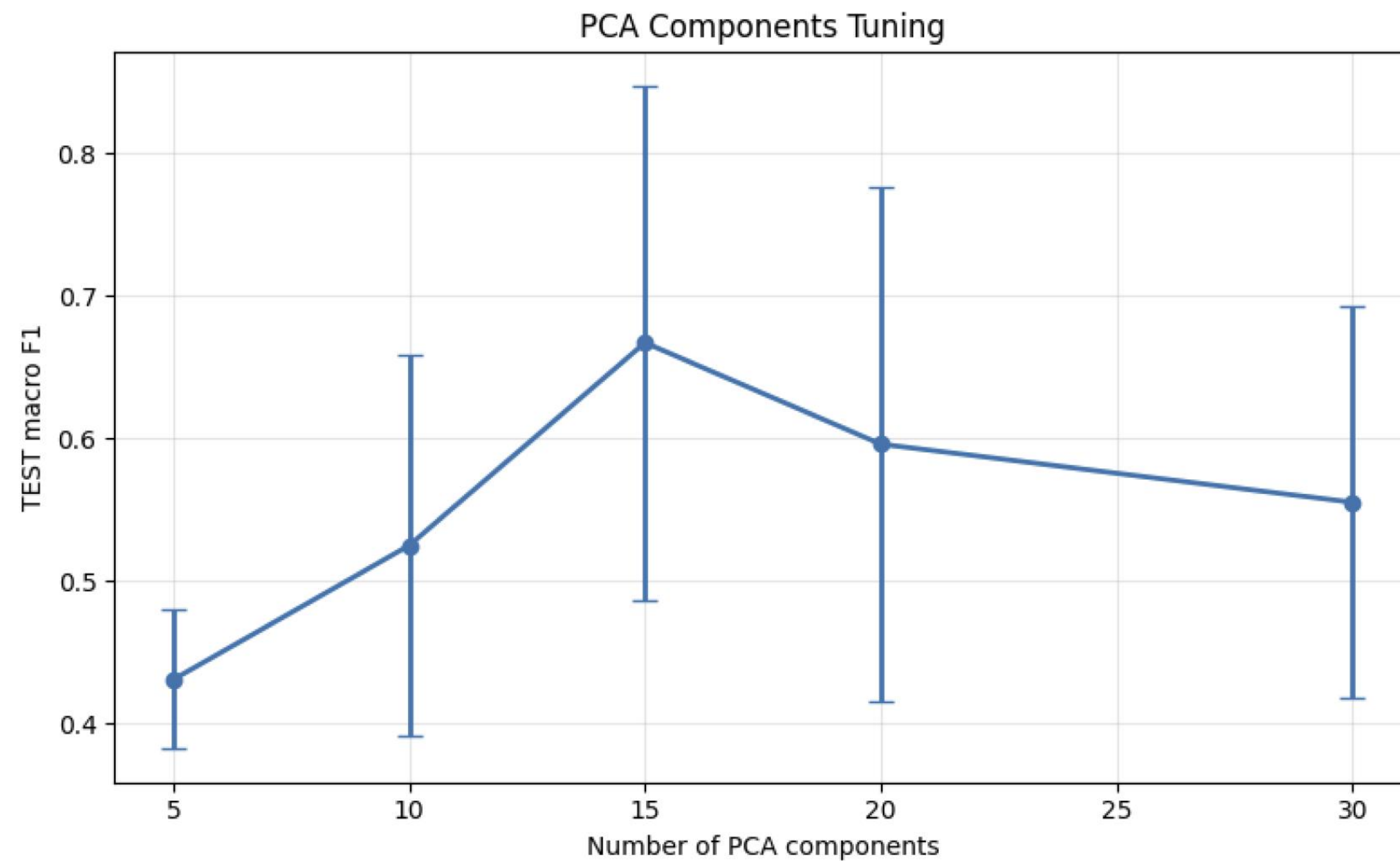
Instead of directly predicting the three classes, we trained two threshold-based classifiers:

- $AHI \geq 5$: apnea vs healthy
- $AHI \geq 30$: severe vs non-severe

This approach tries to reduce the confusion between adjacent severity levels.

→ We obtain a slight improvement, but the problem is still too complex. Let's try to change the input space.

Multi-Modal Logistic Classifier

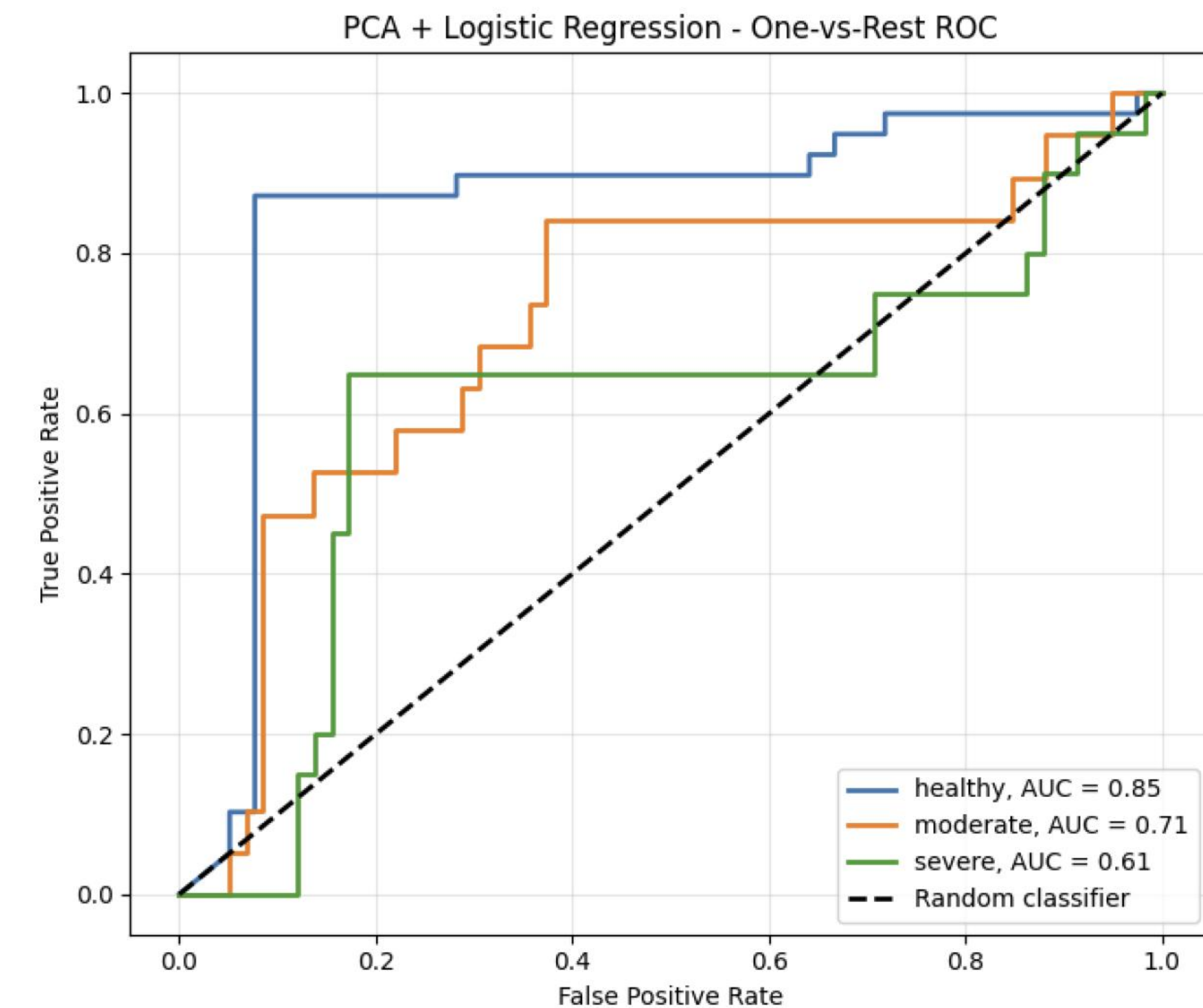
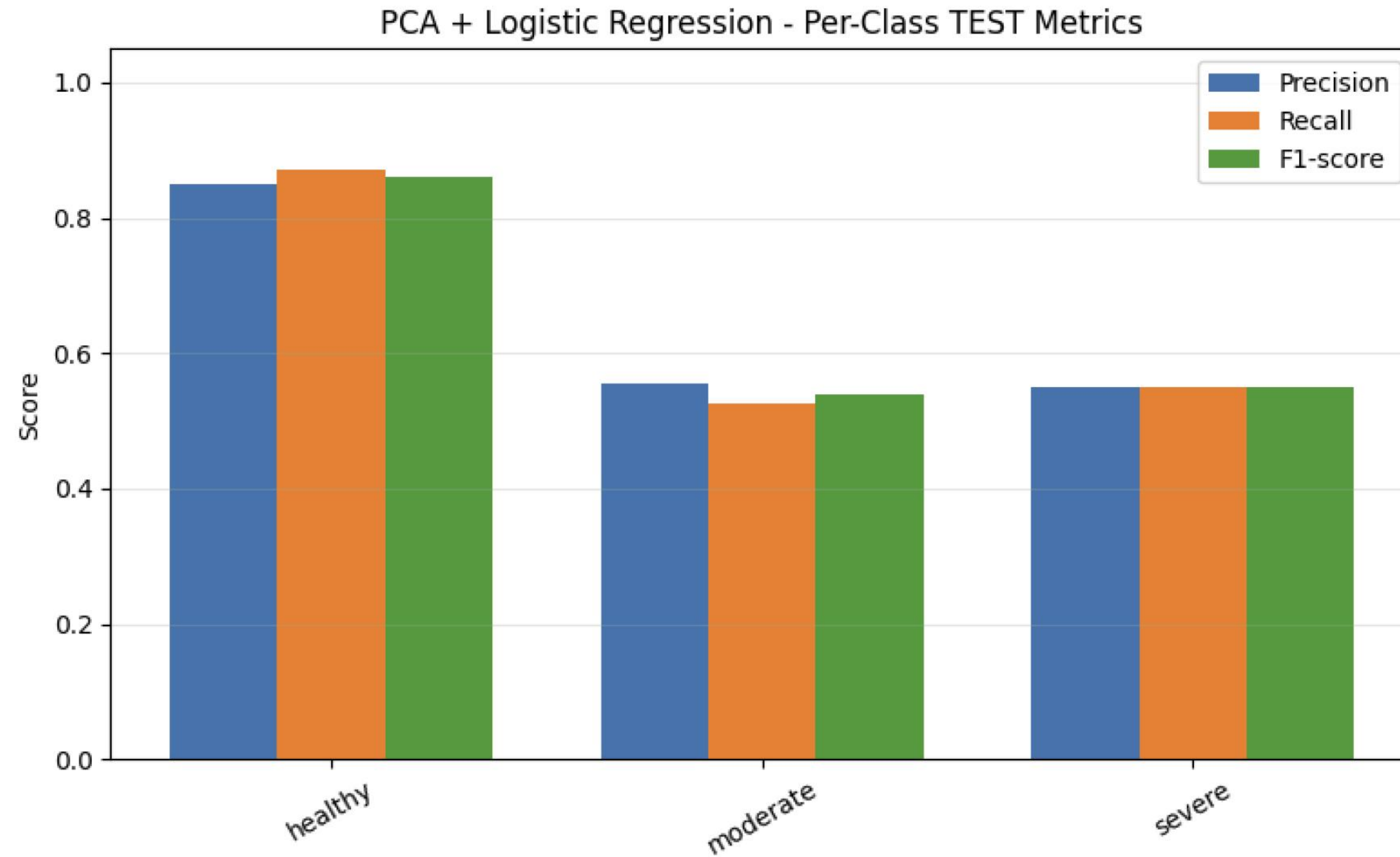


The problem: The input space is extremely high-dimensional, with many input features being extremely correlated (temporal and spatial dependency for aggregate values representing channels and time).

Let's try to perform PCA before applying Logistic Regression to reduce the input dimensionality

→ Cross-validation (on the training set only) suggests N=15

Multi-Modal Logistic Classifier

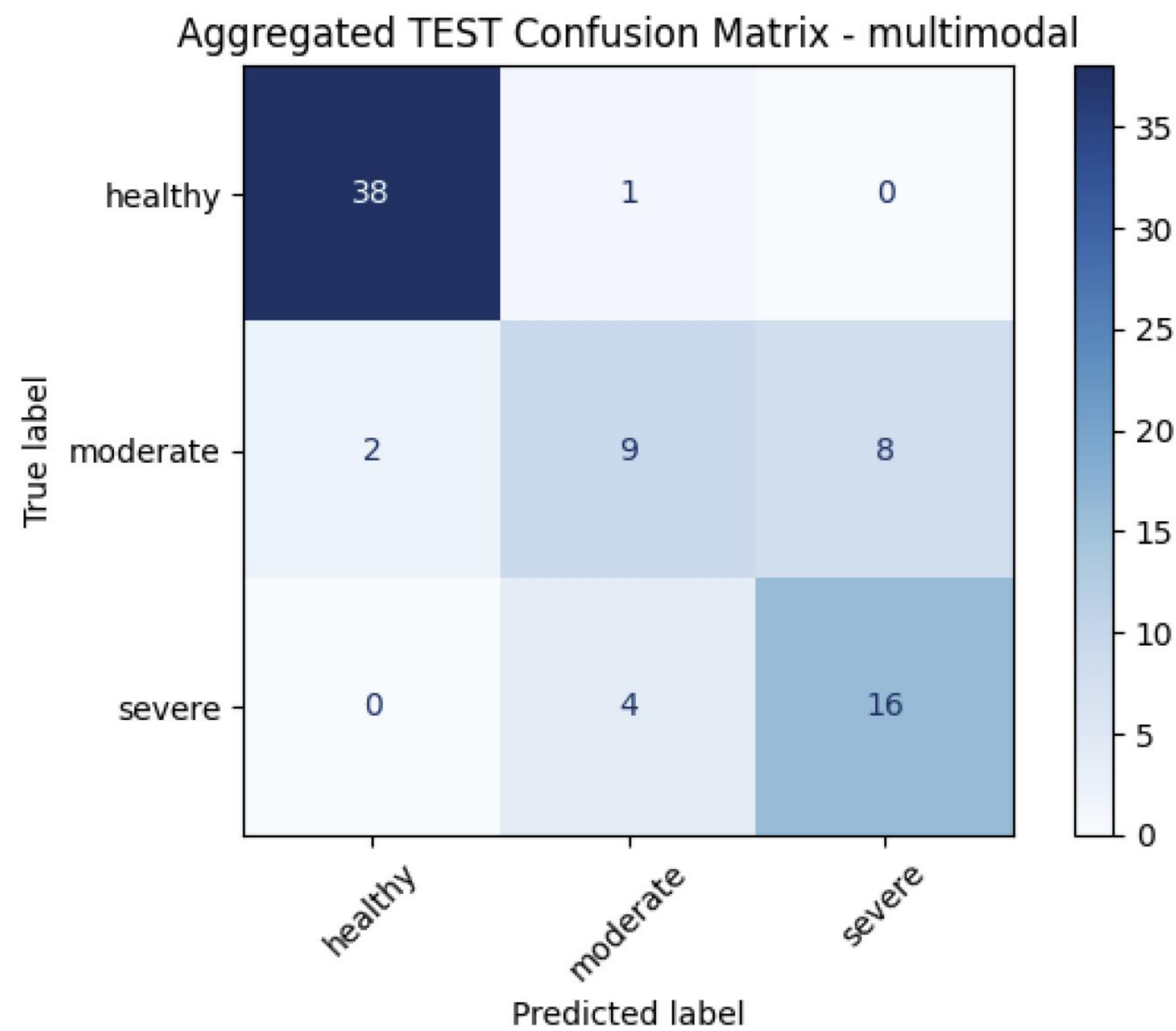


The performance for “Moderate” samples increases at the cost of the other two

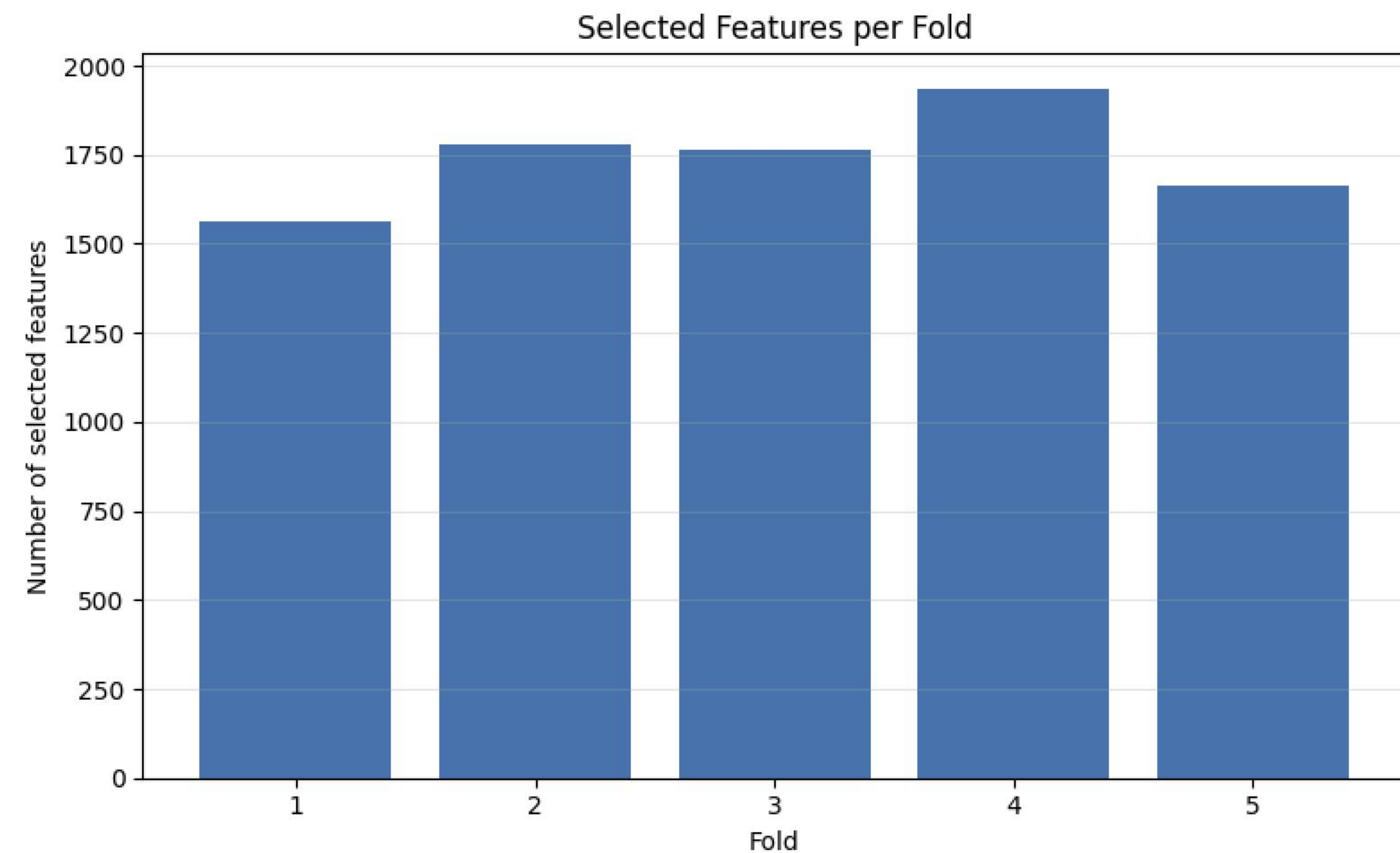
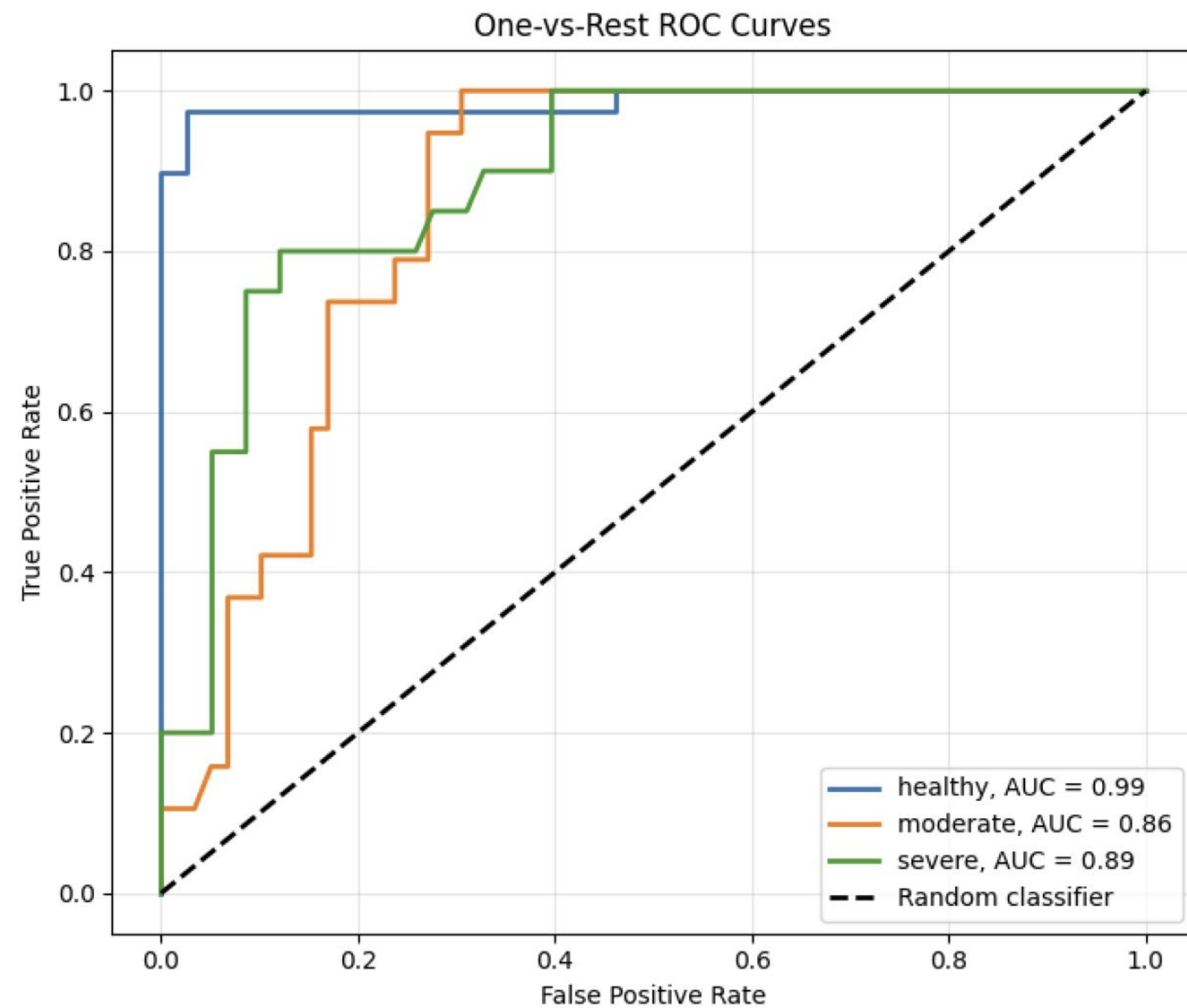
- As we can see in the K-fold (surrogate) uncertainty, the model has extremely high variance probably due to multicollinearity and the low amount of data.

→ Let's try to use a model which is more robust to variance to try and average it out,
Random Forests

Random Forest Classifier



Random Forest Classifier

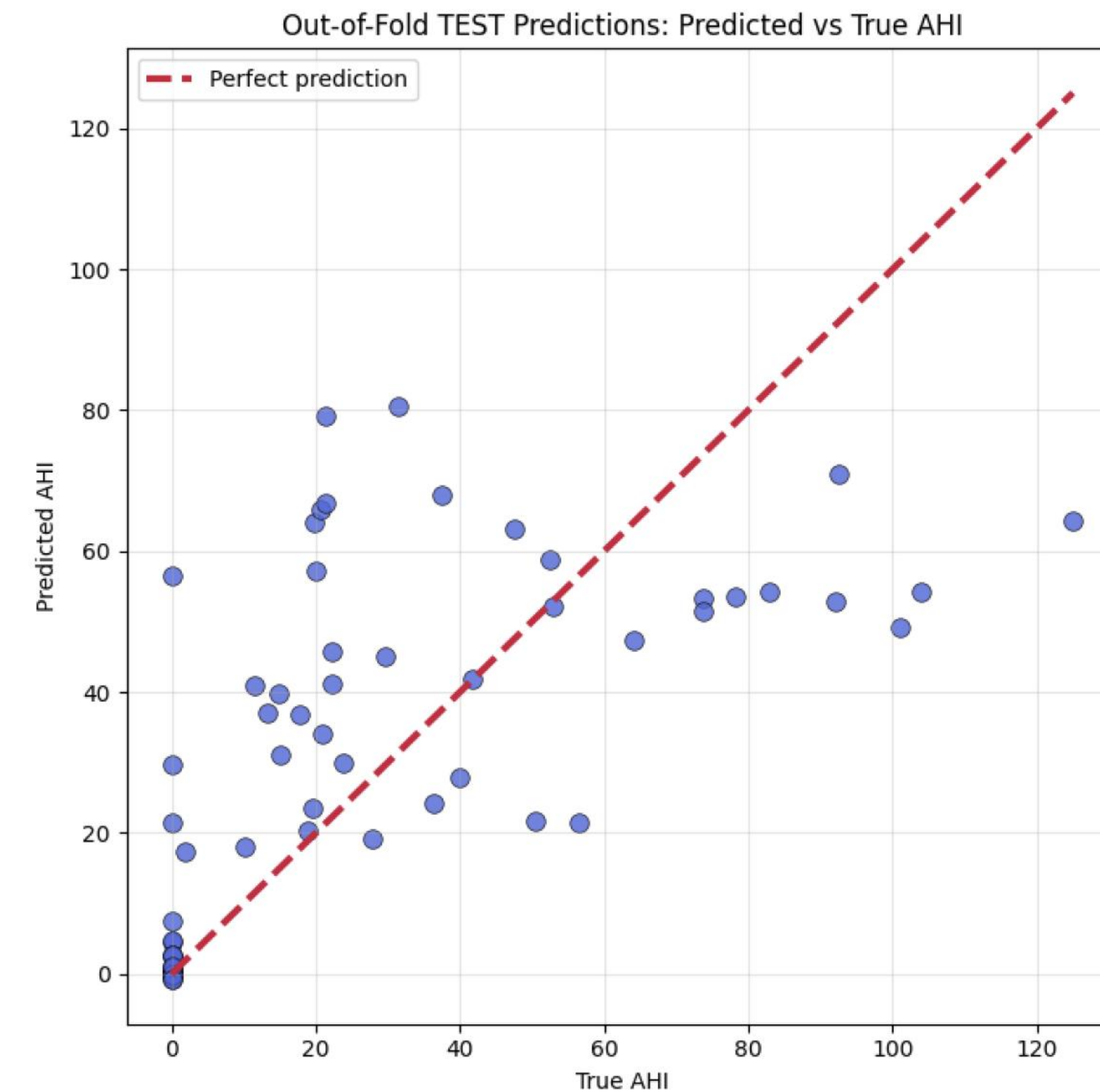
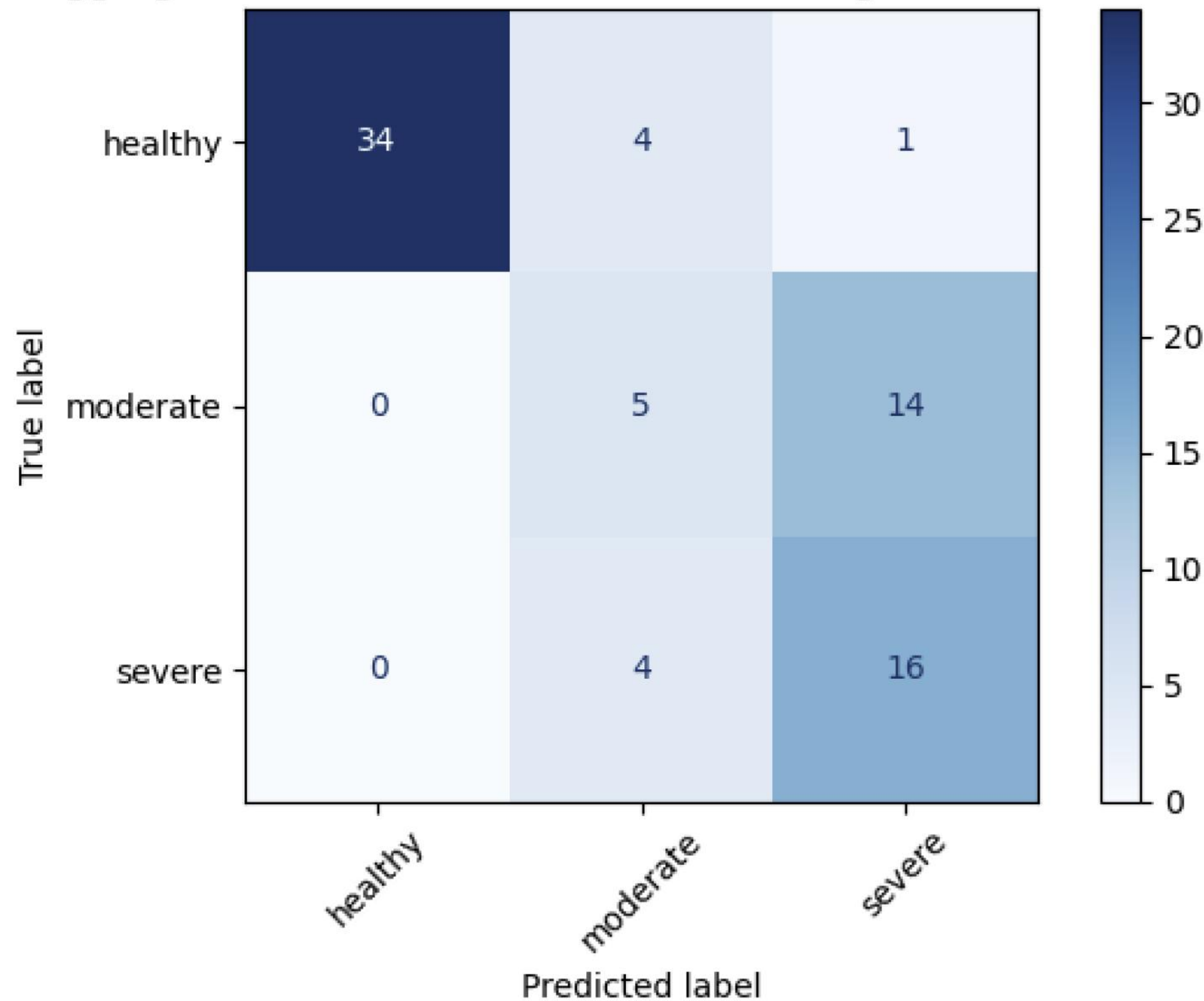


- Cross validation (again, on the training set) to analyze important features
- Healthy subjects are recognized very well. Most errors occur between moderate and severe classes: the moderate class remains the hardest to classify.
- Many features are selected, suggesting that the useful signal is distributed across the EEG embedding (attacks can happen at any point in the night)

Let's try with a
different approach:
Regression

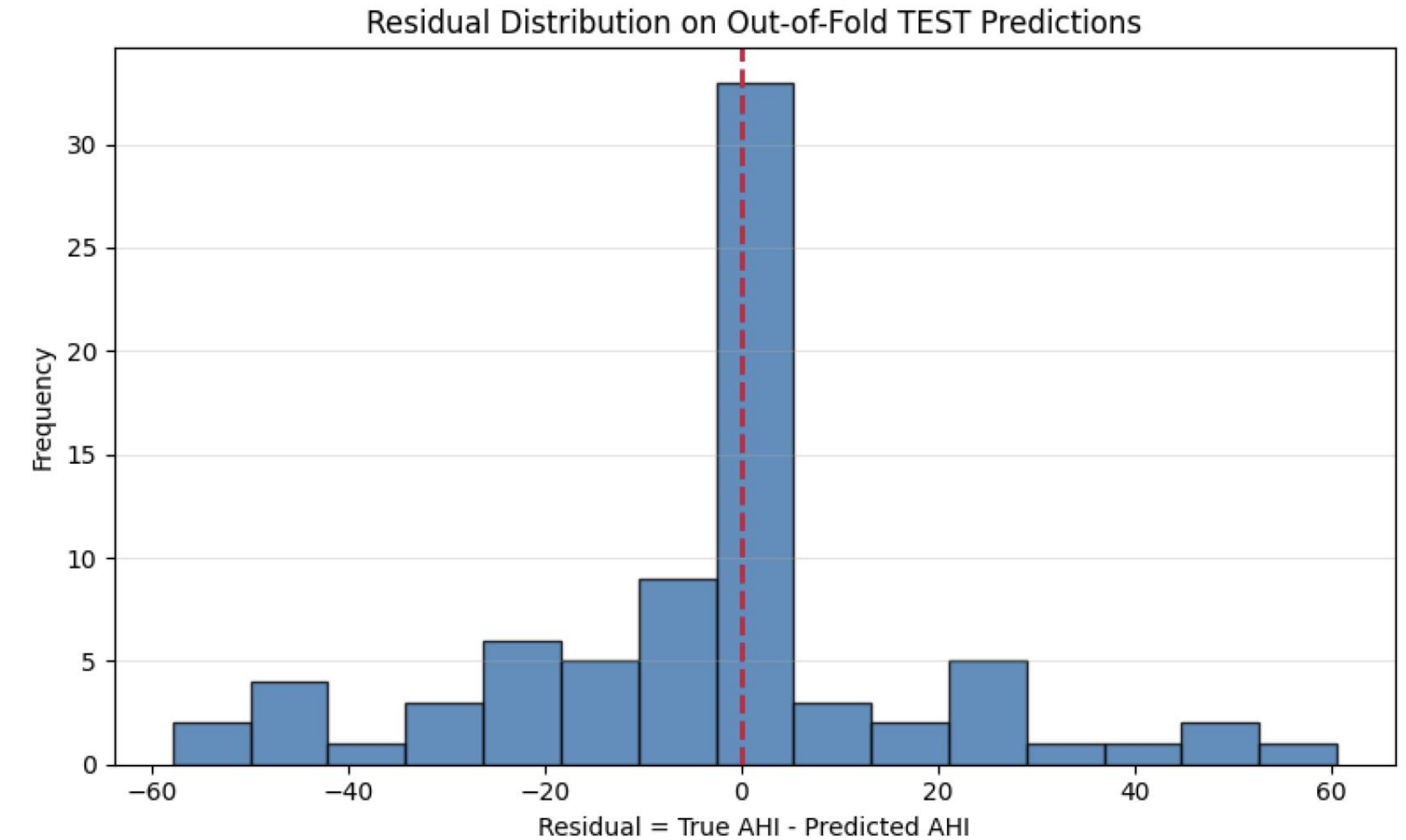
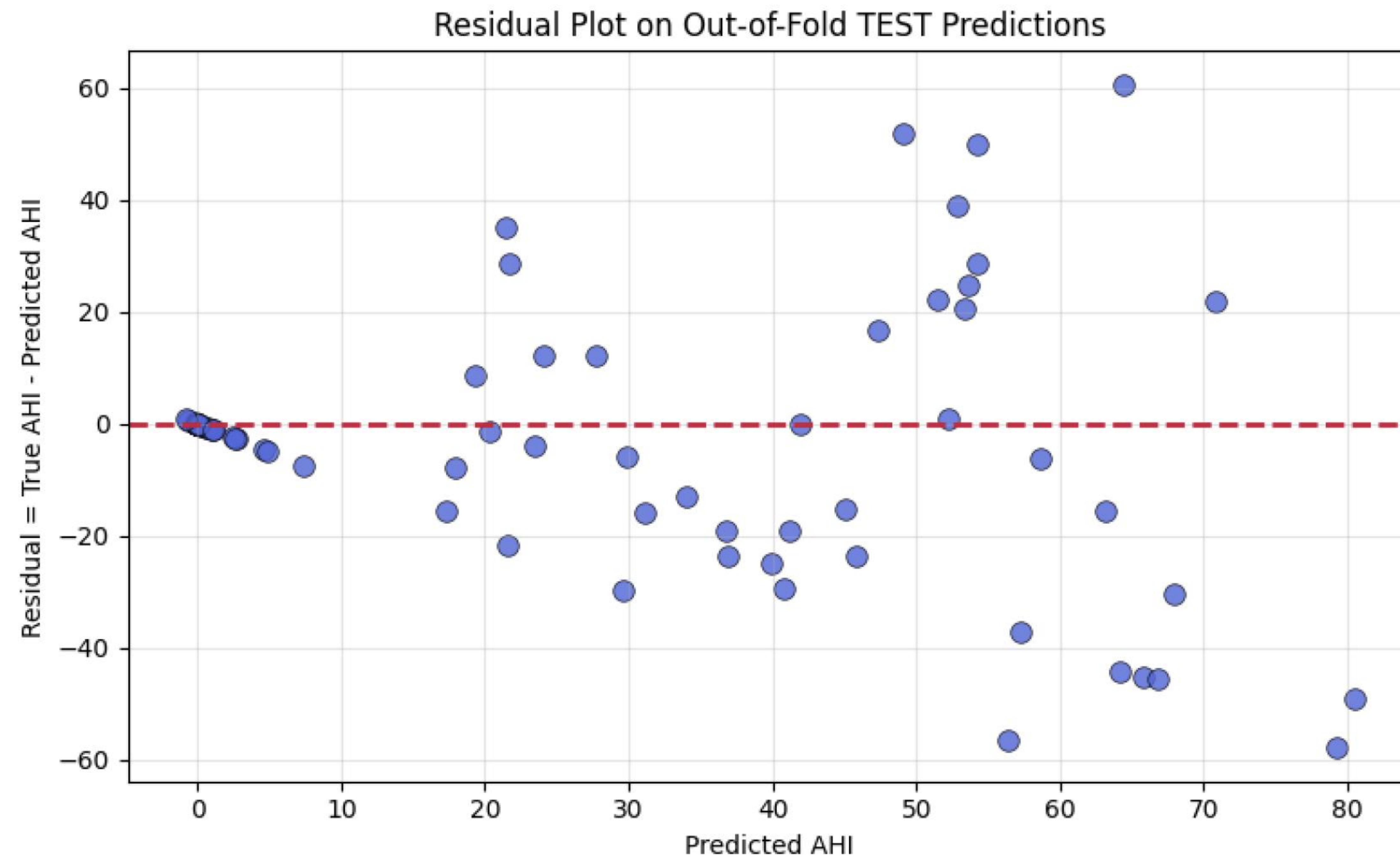
Gradient Boosting Regression-To-Severity

Aggregated TEST Confusion Matrix: GB Regression -> Severity



- Healthy subjects are classified well.
 - Moderate cases are often predicted as severe.
 - High AHI values are underestimated.
- This suggests that regression-to-threshold is unstable around severity boundaries.

Gradient Boosting Regression-To-Severity



- Residuals are widely spread, especially for pathological subjects.
- Positive residuals indicate underestimated AHI, while negative residuals indicate overestimated AHI.
- The model has no strong global bias, but individual predictions are unstable.

→ The normal distribution is trivially not a good model for residuals, we are training with almost half the points having $y=0.0$

Can we do much better?

EEG signals have structure that gets destroyed when flattened into a single embedding vector and fed to a linear or tree-based model

- Temporal dependencies: Each channel is a time series. A flat vector treats every time point as an independent, exchangeable feature, discarding dynamics.
- Spatial dependencies: Electrodes are arranged on the scalp with known positions. Flattening ignores which channels are near each other or functionally connected.
- The models we have tried are permutation invariant with respect to predictors. But EEG features are not exchangeable: position in time and space carries meaning.
- Local vs. global patterns: Apnea-related events (desaturations, EEG arousals, micro-arousals) are localized in time.

What's more natural: Architectures that respect the data's geometry and temporal structure, handling windows naturally

5. Neural Architectures

Discussion

The modern paradigm:

- A single network learns a shared representation, with task-specific output heads
- Regression head: linear output, trained with MSE on AHI
- Classification head: softmax output, trained with cross-entropy on severity class
- Both can be trained jointly or the backbone frozen and heads swapped post-hoc

Why this is appealing?

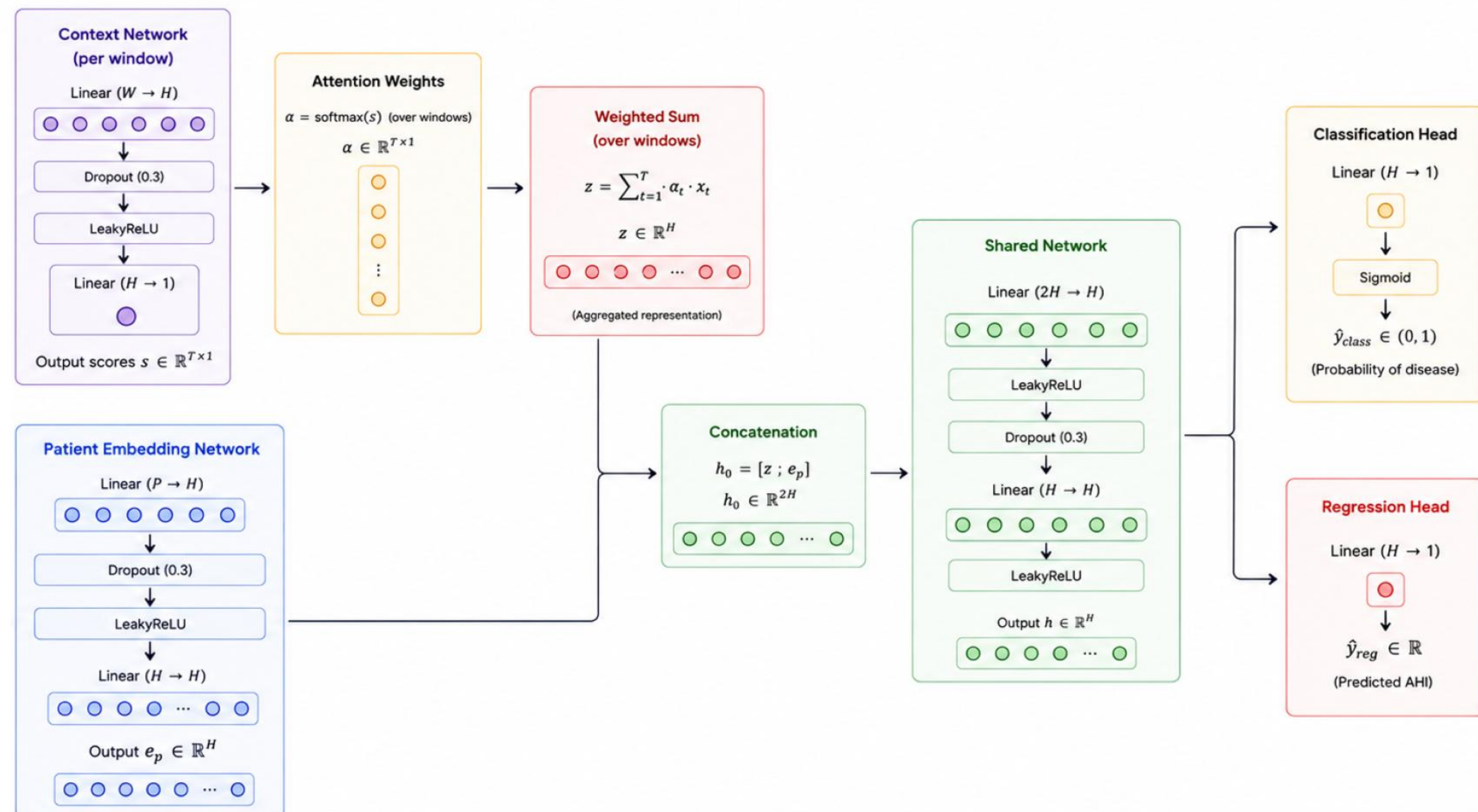
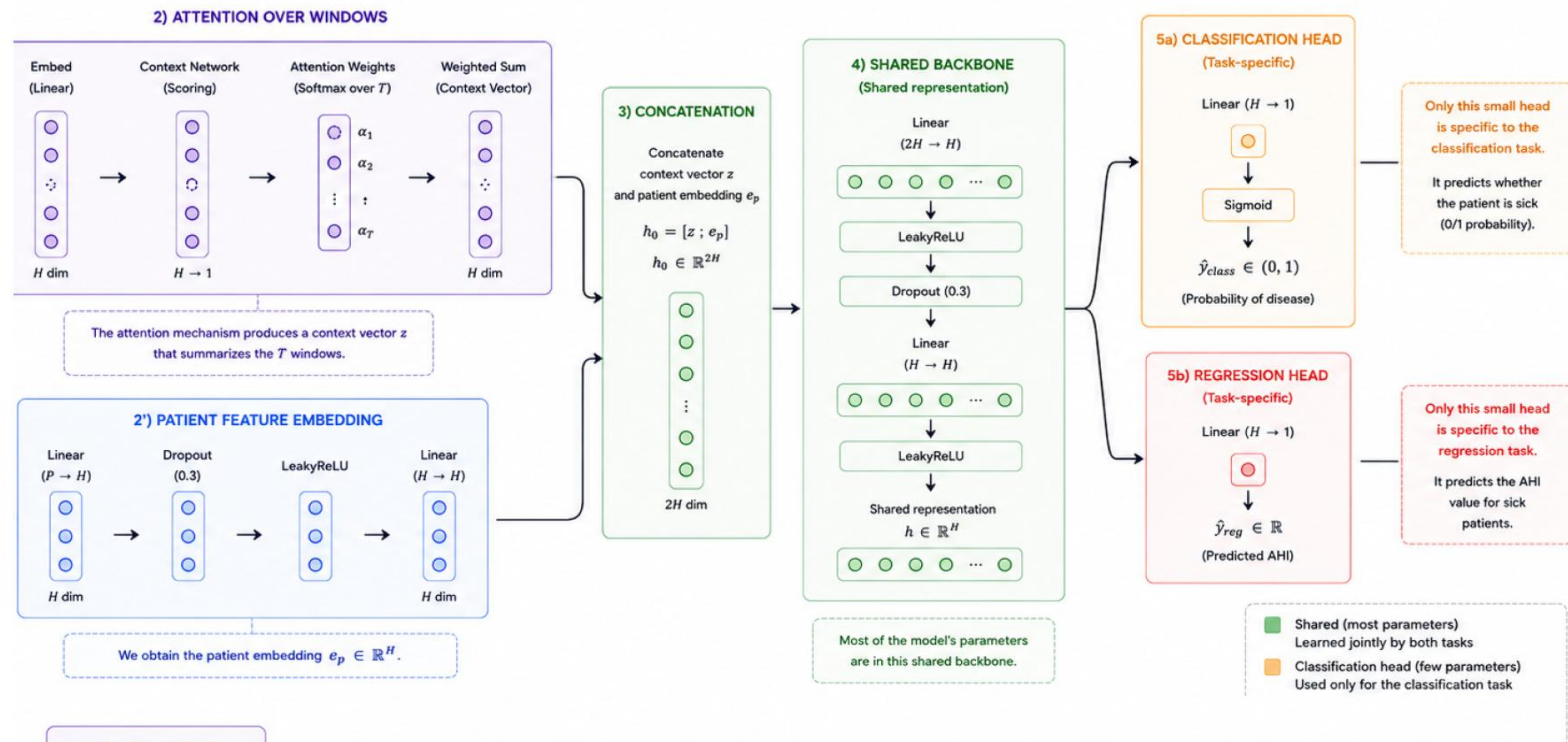
- No need to choose between regression and classification at architecture level
- The backbone can leverage temporal dependencies natively (e.g. LSTM, Transformer)
- Models with attention or RNNs *reason directly over windows of EEG without manual feature extraction*

Cons

- ~78 samples is far below what these models reliably need
- Architecture and hyperparameter search is non-trivial

Architectures

Architecture: Shared Representation with Task-Specific Heads



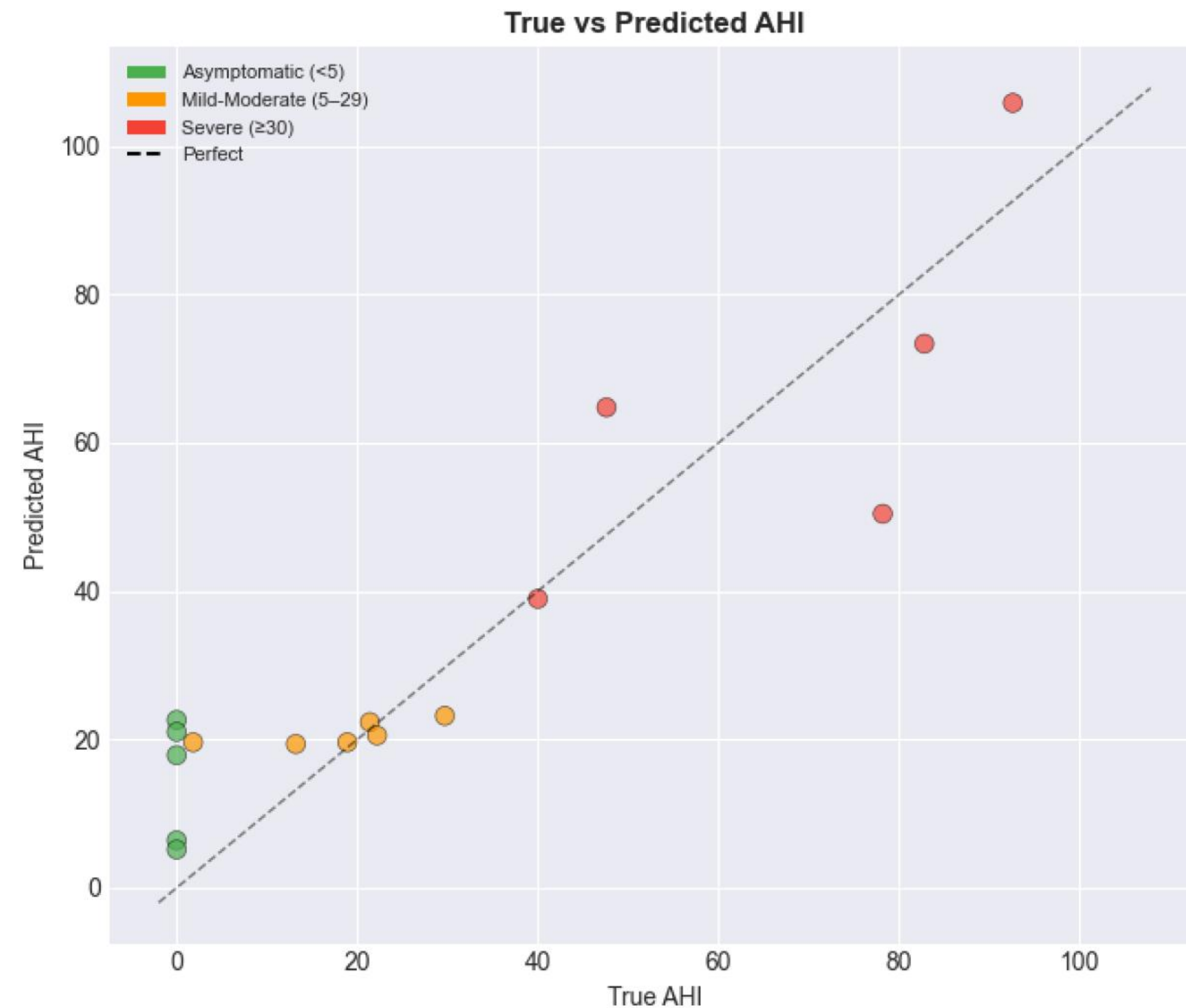
1. Shared-Core MLP with Dual Head

- Input is the flattened feature matrix across windows
- A shared MLP backbone (Linear + ReLU + Dropout blocks) learns a common patient-level representation
- Heads can be trained jointly with a combined loss or independently

2. Context-Attention Network with Dual Head

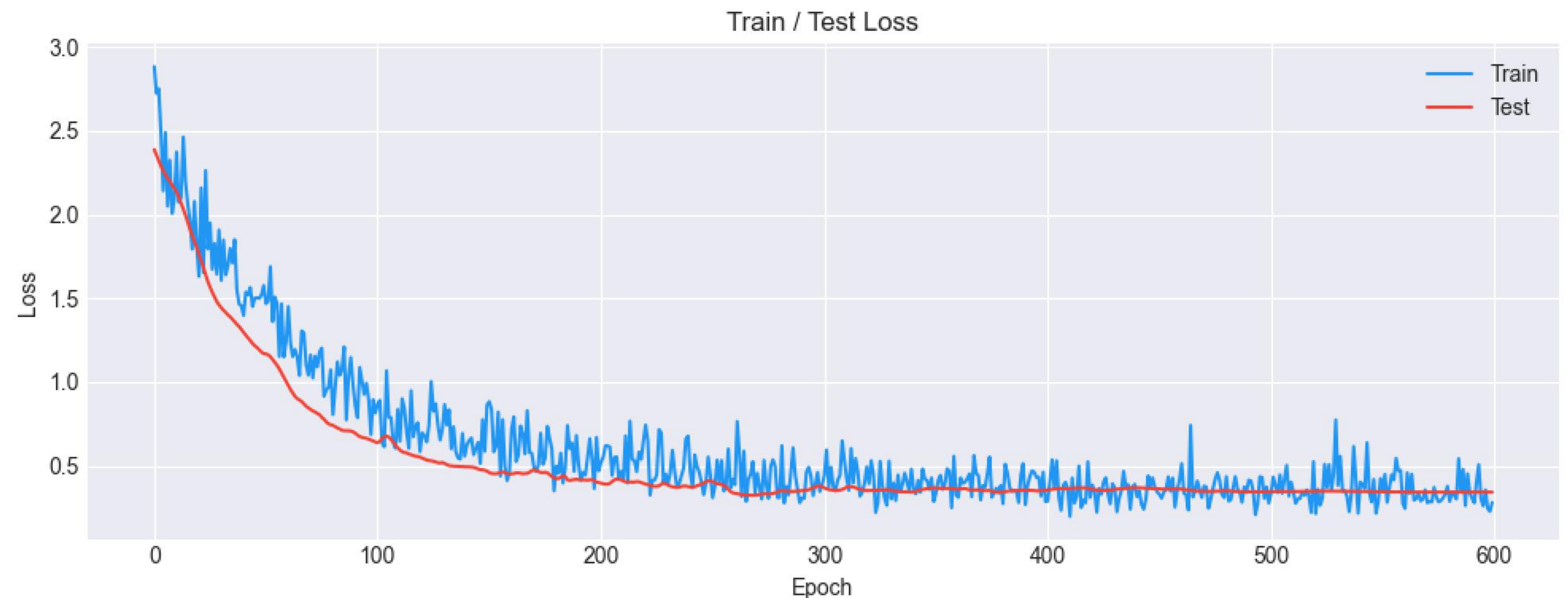
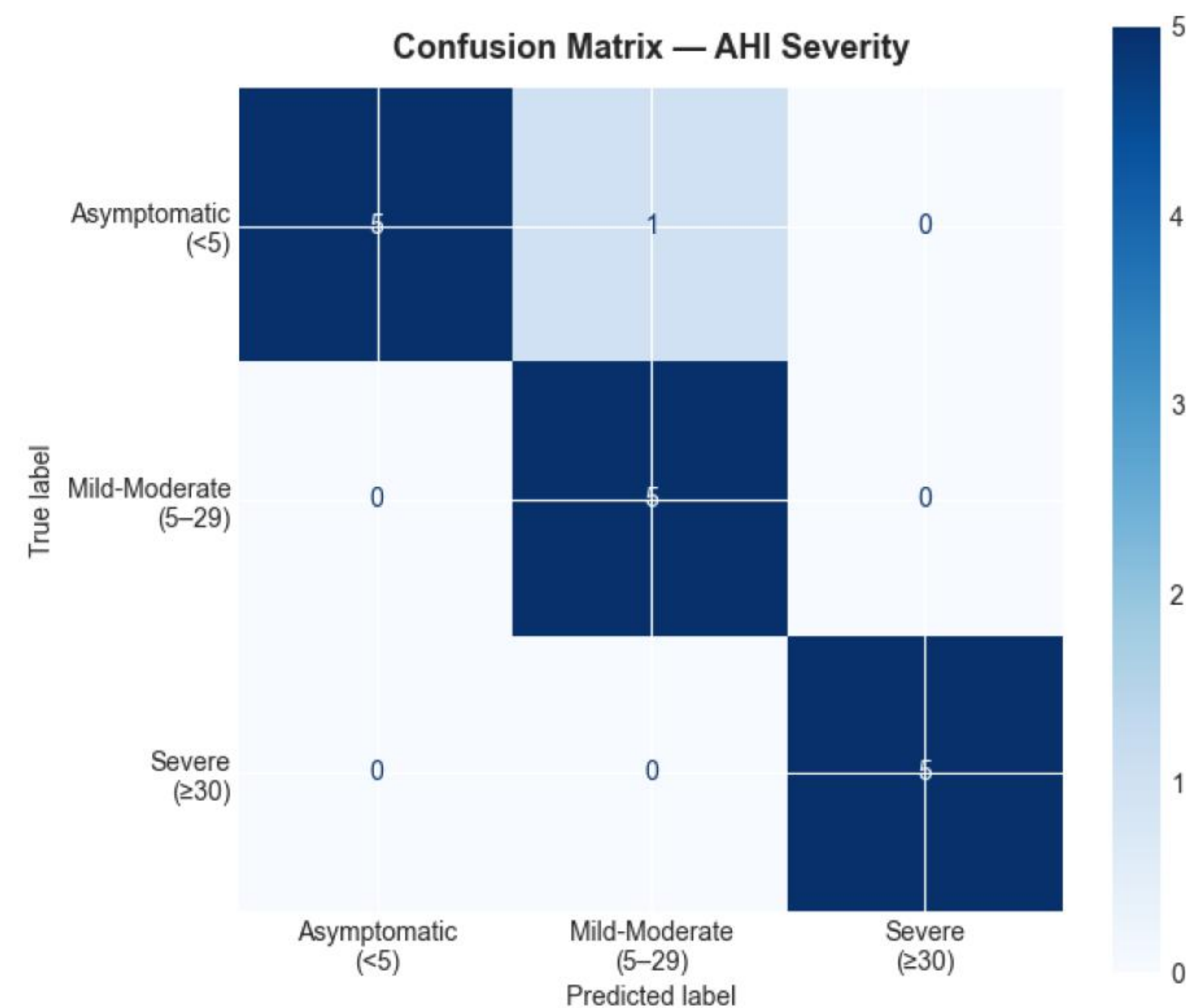
- A context vector is obtained as the weighted sum of window embeddings
- Attention weights are interpretable: they highlight which portions of the night the model relies on most, potentially aligning with clinically meaningful epochs such as REM onset or apnea clusters

Neural Networks



- L1 regularization & dropout as embedded feature selection
- Nested k-fold (Adam optimizer):
 - inner loop tunes alpha parameter for L1 loss
 - Outer loop evaluates performance (precision, recall, accuracy)

→ Problem: data is scarce, with small batches we increase variance during SGD, which large batches we have a catastrophic averaging which collapses the model to predict an average



6. Gaining insight from global context

Network Saliency

- Extracted from attention weights in attention-based neural architectures
- A value for each time window in EEG recording: attention weight quantifies how much the model "values" that window
- Weight directly multiplies the learned embedding representation, so that

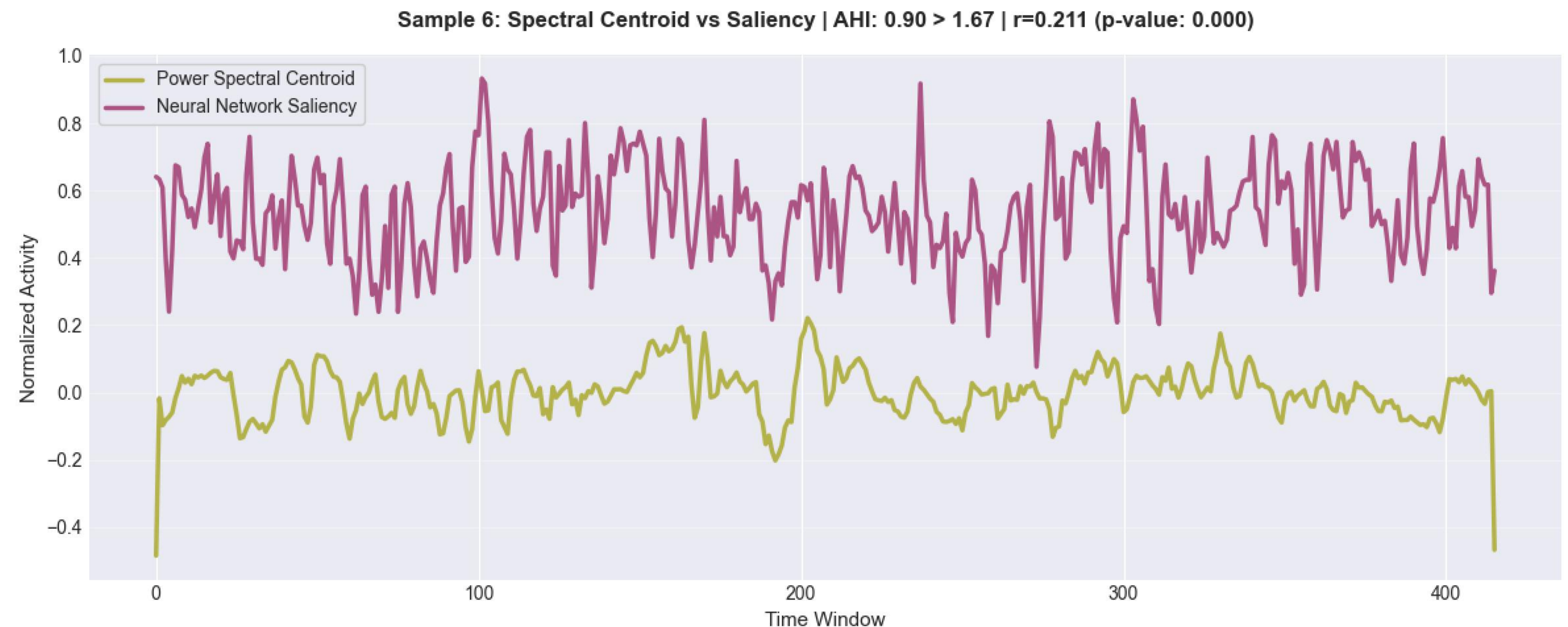
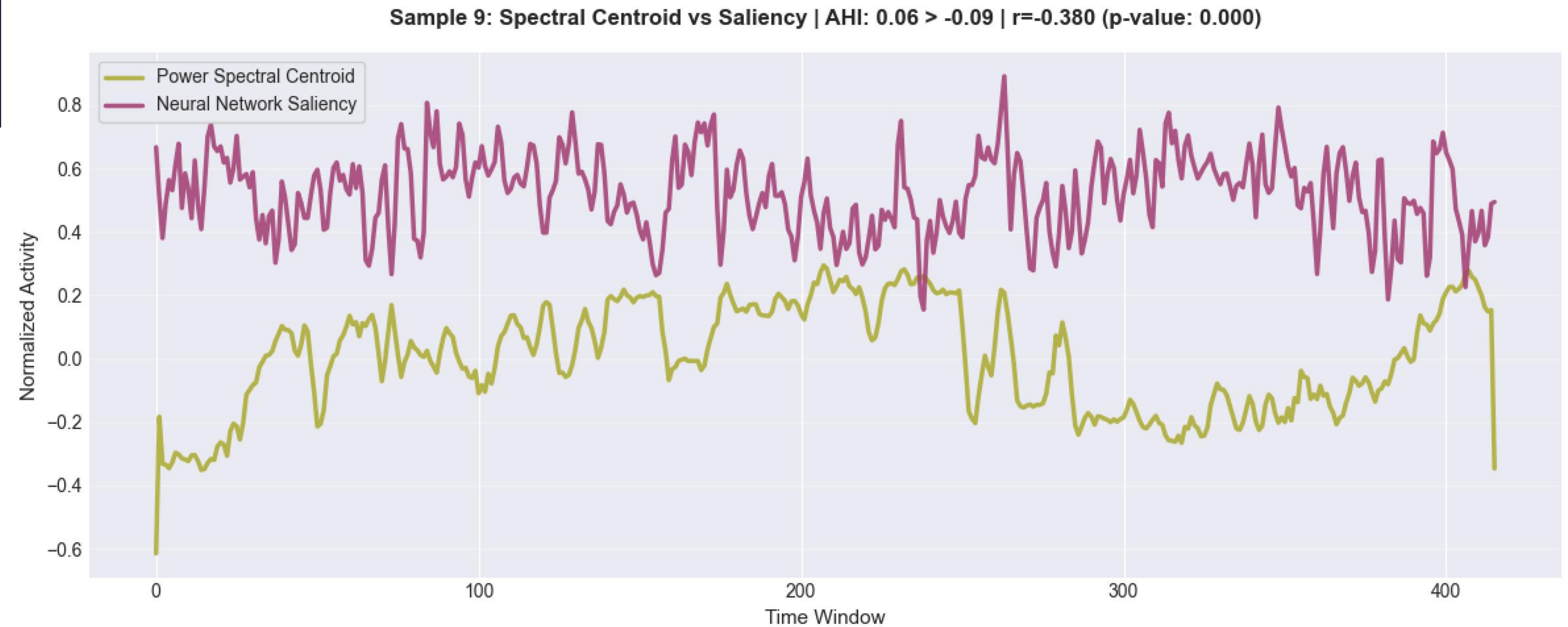
Windows with high saliency contain diagnostic information the network uses for its decision

- e.g. saliency = 0 → the embedding was completely deleted from the network
- saliency = 1 → the embedding was completely taken into account in the network

Spectral Centroid

- "Center of mass" of the power spectrum weighted on the relative power intensity
- Low centroid (5 Hz) = dominance of slow waves
- High centroid (15+ Hz) = shifted energy toward fast bands

Centroid increases during/around apnea events



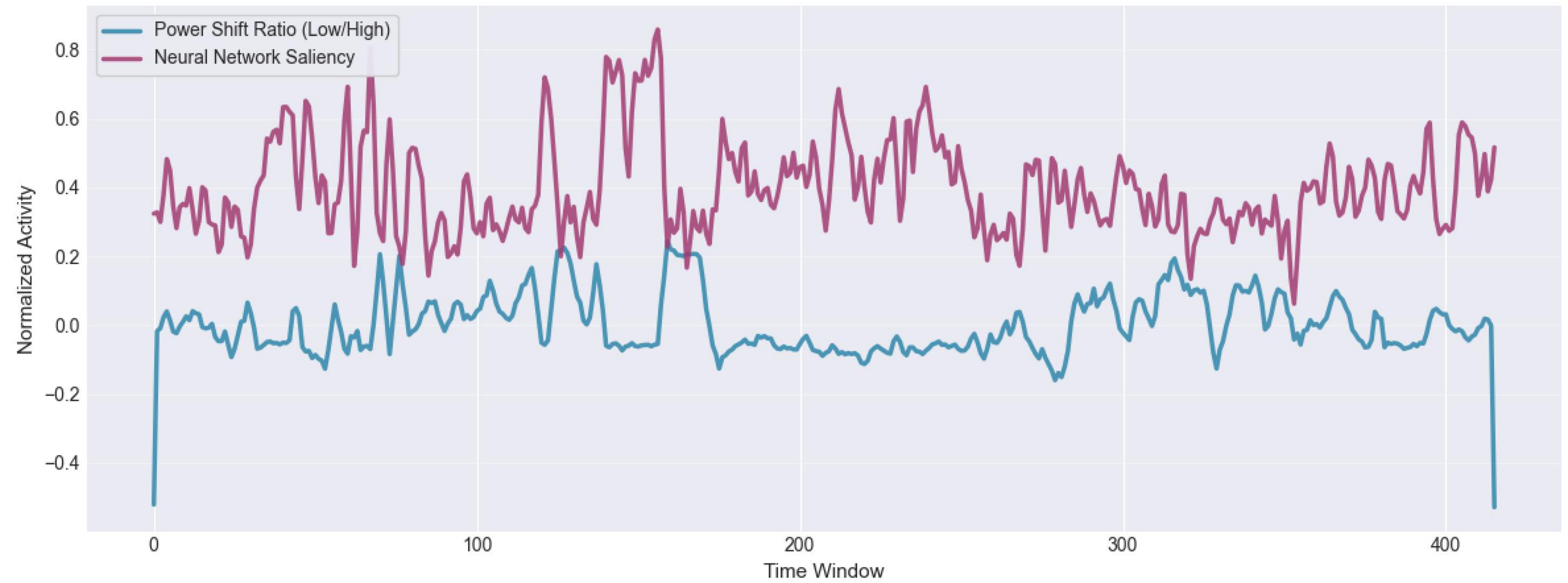
Power Ratio Shift

- Low-band power (0.5–4 Hz) divided by high-band power (12–40 Hz)
- Ratio < 1 = energy shifted to high bands (abnormal, apnea-associated)

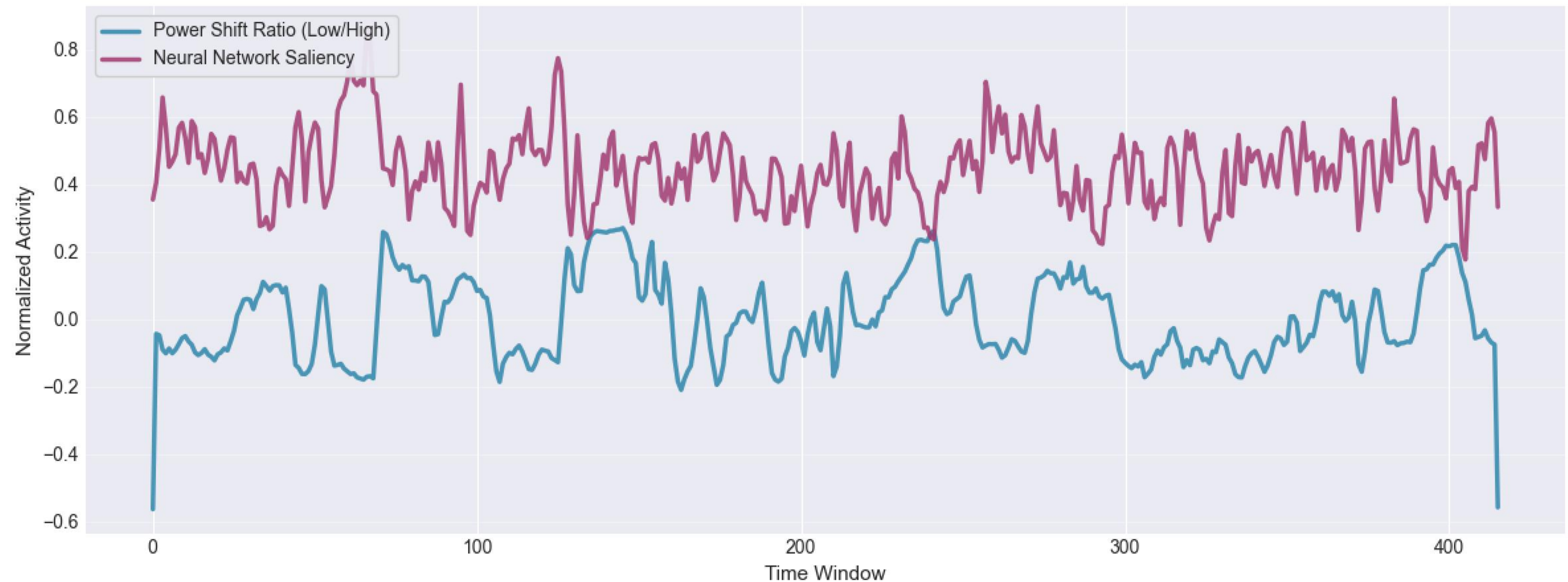
NOTE: The results were normalized for visualization

Validates that attention weights target physically meaningful apnea signatures

Sample 9: Power Shift vs Saliency | AHI: 0.06 > 0.19 | $r=-0.519$



Sample 11: Power Shift vs Saliency | AHI: 0.31 > 0.24 | $r=-0.326$



Thanks for the attention!

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[2] Methods of EEG Signal Features Extraction Using Linear Analysis in Frequency and Time-Frequency Domains - Amjed S. Al-Fahoum, Ausilah A. Al-Fraihat